



Evidence of Photosphere Emission Origin for Gamma-Ray Burst Prompt Emission

Yan-Zhi Meng^{1,2} ¹ School of Astronomy and Space Science, Nanjing University, Nanjing 210023, People's Republic of China; yzmeng@nju.edu.cn² Key Laboratory of Modern Astronomy and Astrophysics (Nanjing University), Nanjing 210023, Ministry of Education, People's Republic of China

Received 2022 March 18; revised 2022 September 21; accepted 2022 September 30; published 2022 December 7

Abstract

The physical origins of gamma-ray burst (GRB) prompt emission (photosphere or synchrotron) are still subject to debate, after more than five decades. Here, we find that many of the observed characteristics of 15 long GRBs, which have the highest prompt emission efficiency ϵ_γ ($\epsilon_\gamma \gtrsim 80\%$), strongly support photosphere (thermal) emission origin, in the following ways: (1) the relation between E_p and E_{iso} is almost $E_p \propto (E_{\text{iso}})^{1/4}$ and the dispersion is quite small; (2) the simple power-law shape of the X-ray afterglow light curves and the presence of significant reverse shock signals in the optical afterglow light curves; (3) the best fits using the cutoff power-law model for time-integrated spectra; and (4) the consistent efficiency from observations (with E_{iso}/E_k) and the predictions from the photosphere emission model (with η/Γ). We then further investigate the characteristics of the long GRBs for two distinguished samples ($\epsilon_\gamma \gtrsim 50\%$ and $\epsilon_\gamma \lesssim 50\%$). It is found that the different distributions for E_p and E_{iso} , and the similar observed efficiency (from the X-ray afterglow) and theoretically predicted efficiency (from the prompt emission or the optical afterglow), follow the predictions of the photosphere emission model well. Also, based on the same efficiency, we derive an excellent correlation of $\Gamma \propto E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4}$ to estimate Γ . Finally, we show that different distributions for E_p and E_{iso} , and the consistent efficiency, exist for short GRBs. We also give a natural explanation of the extended emission ($\epsilon_\gamma \lesssim 50\%$) and the main pulse ($\epsilon_\gamma \gtrsim 50\%$).

Unified Astronomy Thesaurus concepts: [Gamma-ray bursts \(629\)](#)

1. Introduction

More than 50 yr after its discovery, the radiation mechanism of gamma-ray burst (GRB) prompt emission (photosphere emission or synchrotron emission) still remains unidentified (e.g., Mészáros 2002; Zhang & Yan 2011; Uhm & Zhang 2014; Zhang et al. 2018a, 2018b; Geng et al. 2018; Lin et al. 2018; Li et al. 2019; Geng et al. 2019; Burgess et al. 2020; Yang et al. 2020; Zhang 2020; Iyyani & Sharma 2021; Li et al. 2021; Vyas et al. 2021; Zhang et al. 2021, 2021; Vereshchagin et al. 2022). The photospheric emission is the basic prediction of the classical fireball model (Goodman 1986; Paczynski 1986) for a GRB, because the optical depth τ at the jet base is much larger than unity (e.g., Piran 1999). As the fireball expands, the optical depth drops down. The internally trapped thermal photons finally escape at the photosphere ($\tau = 1$). Indeed, based on spectral analysis, a quasi-thermal component has been found in several Swift GRBs (Ryde 2004, 2005; Ryde & Pe'er 2009) and Fermi GRBs (Guiriec et al. 2011; Axelsson et al. 2012; Ghirlanda et al. 2013; Guiriec et al. 2013; Larsson et al. 2015; Tang et al. 2021; Deng et al. 2022; Wang et al. 2022; Zhao et al. 2022), especially GRB 090902B (Abdo et al. 2009; Ryde et al. 2010; Zhang 2011). But whether the typically observed Band function (a smoothly joint broken power law; Band et al. 1993) or a cutoff power law (CPL) can be explained by the photosphere emission—namely, by the photospheric emission model—remains unknown (e.g., Abramowicz et al. 1991; Thompson 1994; Mészáros & Rees 2000; Rees & Mészáros 2005; Pe'er & Ryde 2011; Fan et al. 2012; Lazzati et al. 2013; Ruffini et al. 2013; Bégué & Pe'er 2015; Gao et al. 2015; Pe'er et al. 2015; Ryde et al. 2017; Acuner & Ryde 2018;

Hou et al. 2018; Meng et al. 2018; Li 2019a, 2019b; Meng et al. 2019; Acuner et al. 2020; Dereli-Bégué et al. 2020; Li 2020; Vereshchagin & Siutsou 2020; Wang et al. 2020; Meng et al. 2022; Parsotan & Lazzati 2022; Song & Meng 2022). If this scenario is true, the quasi-thermal spectrum should be broadened. Theoretically, two different broadening mechanisms have been proposed (see Appendix A): subphotospheric dissipation (namely, the dissipative photosphere model; Rees & Mészáros 2005; Giannios & Spruit 2007; Vurm & Beloborodov 2016; Beloborodov 2017; Bhattacharya et al. 2018; Bhattacharya & Kumar 2020) or geometric broadening (namely, the probability photosphere model; Pe'er 2008; Pe'er & Ryde 2011; Lundman et al. 2013; Deng & Zhang 2014; Meng et al. 2018, 2019, 2022).

Previously, some of the implications of the statistical properties of the spectral analysis results for a large GRB sample have appeared to support the photosphere emission model. First, lots of bursts have a low-energy spectral index α that is harder than the death line (or the maximum value, $\alpha = -2/3$) of the basic synchrotron model, especially for short GRBs and the peak-flux spectrum (Kaneko et al. 2006; Zhang et al. 2011; Burgess et al. 2017). Second, the spectral width is found to be quite narrow for a significant fraction of GRBs (Axelsson & Borgonovo 2015; Yu et al. 2015). Third, for half or more of GRBs, the CPL is the best-fit empirical model (Goldstein et al. 2012; Gruber et al. 2014; Yu et al. 2016), indicating that the photosphere emission model can naturally interpret their high-energy spectra. Here, we find more convincing evidence for the photosphere emission origin (especially the probability photosphere model) of GRB prompt emission, by obtaining the prompt efficiency ϵ_γ .

The paper is organized as follows. In Section 2, we state the data accumulation and the scaling relations predicted by the photosphere model. In Section 3, we describe the evidence from long GRBs with extremely high prompt efficiency ϵ_γ .



Original content from this work may be used under the terms of the [Creative Commons Attribution 4.0 licence](#). Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.

Table 1
The Observed Quantities for Bursts with Extremely High Efficiency ($\epsilon_\gamma \gtrsim 80\%$)

GRB	z	L_{iso} (10^{52} erg s $^{-1}$)	E_{iso} (10^{52} erg)	$L_{\text{X},11}$ (10^{45} erg s $^{-1}$)	Efficiency	$E_{p,z}$ (keV)	α	β	Best-fitted Model
990705 ^a	0.8424	1.61 ± 0.15^c	21.8 ± 0.8	1.14	0.83 (0.99 ^f)	551 ± 17	-0.72 ± 0.03	-2.68 ± 0.15^e	
000210 ^a	0.8463	9.76 ± 0.5^c	19.3 ± 0.5	1.67	0.74 (0.97 ^f)	687^{+39}_{-37}			
060927 ^b	5.47	10.8 ± 0.8	7.56 ± 0.46	0.28	0.96	459 ± 90	-0.81 ± 0.36		
061007 ^b	1.261	10.9 ± 0.9	101 ± 1.4	1.11	0.97	965 ± 27	-0.75 ± 0.02	-2.79 ± 0.09	
080319B ^b	0.9382	10.2 ± 0.9	142 ± 3	3.90	0.91	1307 ± 43	-0.86 ± 0.01	-3.59 ± 0.45	
080607 ^b	3.0363	225.9 ± 45.3	186 ± 10	3.46	0.97	1691 ± 169	-1.08 ± 0.06		
081203A ^b	2.05	2.82 ± 0.19	35.0 ± 12.8	1.52	0.90	1541 ± 756	-1.29 ± 0.14		
110205A ^b	2.22	2.51 ± 0.34	55.9 ± 5.3	2.12	0.92	715 ± 238	-1.52 ± 0.14		
110818A ^c	3.36	6.76 ± 0.76	21.7 ± 1.02	0.99	0.93	799.45 ± 371.90	-1.19 ± 0.08		CPL ^d
120729A ^a	0.80		2.3 ± 1.5	0.034	0.94	559 ± 57			
130606A ^a	5.913		28.3 ± 5.2	1.26	0.95	2032^{+622}_{-346}	-1.14 ± 0.15		CPL ^e
131108A ^a	2.40	26.22 ± 0.6^c	54 ± 2.4	1.98	0.93	1217^{+105}_{-88}	-1.16 ± 0.07		CPL ^e
150821A ^a	0.755	0.77 ± 0.03^c	15.5 ± 1.2	1.05	0.78	765^{+188}_{-126}	-1.52 ± 0.05		CPL ^e
160410A ^{a,g}	1.717		9.3 ± 1.8	0.44	0.89	3853^{+1429}_{-973}	-0.71 ± 0.20		CPL ^e
161014A ^c	2.823	5.21 ± 0.52	9.49 ± 0.50	0.55	0.90	646.18 ± 55.13	-0.76 ± 0.08		CPL ^d
170214A ^a	2.53	30.32 ± 0.85^c	318.43 ± 0.21	1.94	0.99	1810 ± 34	-0.98 ± 0.01	-2.51 ± 0.10^d	
130606A ^a	5.913		28.3 ± 5.2	1.26	0.95	2032^{+622}_{-346}	-1.14 ± 0.15		CPL ^e

Notes.

^a E_{iso} and $E_{p,z}$ are taken from Minaev & Pozanenko (2020).

^b E_{iso} , $E_{p,z}$, L_{iso} , α , and β are taken from Nava et al. (2012).

^c E_{iso} , $E_{p,z}$ and L_{iso} (or L_{iso} alone) are taken from Xue et al. (2019).

^d The spectral analysis results are from the Fermi GBM Burst Catalog (von Kienlin et al. 2020).

^e The spectral analysis results are from the Konus-Wind Burst Catalog (Tsvetkova et al. 2017).

^f The extremely high efficiency claimed in Lloyd-Ronning & Zhang (2004).

^g The short GRB.

($\epsilon_\gamma \gtrsim 80\%$). Then, in Section 4, the evidence from long GRBs with $\epsilon_\gamma \gtrsim 50\%$ and $\epsilon_\gamma \lesssim 50\%$ is shown. In Section 5, we illustrate the evidence from short GRBs. A brief summary is provided in Section 6.

2. Data Accumulation and the Scaling Relations Expected by the Fireball Model

2.1. Data Accumulation

Generally, the radiation efficiency of the prompt emission ϵ_γ is defined as $E_\gamma/(E_\gamma + E_k)$. Here, E_γ is the radiated energy in the prompt phase and E_k is the remaining kinetic energy in the afterglow phase.

To obtain ϵ_γ , the isotropic energy E_{iso} (namely E_γ) and $L_{\text{X},11\text{h}}$ (the late-time X-ray afterglow luminosity at 11 hr) data should be accumulated for the GRB sample with the redshift z .³ Because of this, $L_{\text{X},11\text{h}}$ is roughly proportional to E_k (see Appendix B.1).

(1) For 46 long bursts before GRB 110213A, we use the $L_{\text{X},11\text{h}}$ data given in D’Avanzo et al. (2012). Also, E_{iso} , the isotropic luminosity L_{iso} , the peak spectral energy in the rest frame $E_{p,z}$ (or E_p), the low-energy spectral index α , and the high-energy spectral index β are taken from Nava et al. (2012).

(2) For 117 long bursts after GRB 110213A (see Tables 1–3, with $\epsilon_\gamma \gtrsim 80\%$, $\epsilon_\gamma \lesssim 50\%$, and $\epsilon_\gamma \gtrsim 50\%$, respectively), we calculate $L_{\text{X},11\text{h}}$ following the method (see Appendix B.1) in D’Avanzo et al. (2012). E_{iso} , L_{iso} , $E_{p,z}$, $T_{90,i}$ (the intrinsic duration in the rest frame, or T_{90}), and z are mainly taken from Minaev & Pozanenko (2020) and Xue et al. (2019).

Our spectral fitting results for the $\epsilon_\gamma \gtrsim 80\%$ sample are given in Table 4. And for several bursts of the $\epsilon_\gamma \lesssim 50\%$ sample, the perfectly consistent observed efficiency (with E_{iso}/E_k) and the theoretically predicted efficiency of the photosphere model (from the prompt emission) are given in Table 5.

In Table 6, for the long-GRB sample with the detection of the peak time of the early optical afterglow (namely T_p or $T_{p,\text{op}}$, to obtain the Lorentz factor of the outflow Γ and then $(R_{\text{ph}}/R_s)^{-2/3}$) in Ghirlanda et al. (2018), the consistent predicted efficiency (from the prompt emission and the optical afterglow) of the photosphere model is provided. Also, E_{iso} , L_{iso} , and $E_{p,z}$ are taken from Ghirlanda et al. (2018). In Table 7, its subsample (nine bursts) with the maximum Γ (for fixed L_{iso} ; according to the photosphere model, $\epsilon_\gamma = 50\%$) is provided.

In Table 8, the short-GRB sample (with $L_{\text{X},11\text{h}}$ derived in this work) is given, along with that possessing extended emission. E_{iso} and $E_{p,z}$ are taken from Minaev & Pozanenko (2020).

E_{iso} is generally estimated by $E_{\text{iso}} = 4\pi D_L^2 S_\gamma / (1 + z)$, where S_γ is the time-integral fluence in the $1 - 10^4$ keV energy range in the rest frame (in units of erg cm $^{-2}$) and D_L is the luminosity distance. L_{iso} is estimated as $L_{\text{iso}} = 4\pi D_L^2 F_p$, where F_p is the peak flux (in units of erg cm $^{-2}$ s $^{-1}$). $T_{90,i}$ is calculated as $T_{90,i} = T_{90,\text{ob}}/(1 + z)$, where $T_{90,\text{ob}}$ is determined by the time range between the epochs when the accumulated net photon counts reach the 5% level and the 95% level. And $E_p = (1 + z) \cdot E_{p,\text{ob}}$, where $E_{p,\text{ob}}$ is determined by the peak energy in the νF_ν spectrum.

Typically, the afterglow peak time T_p is estimated from the optical afterglow peak time $T_{p,\text{op}}$, since the early X-ray afterglow peak can be produced by “internal” mechanisms (such as the prolonged central engine activity) or bright flares.

³ The redshift data are publicly available at <http://www.mpe.mpg.de/jcg/brbgen.html>.

Table 2
The Observed Quantities and Inferred Efficiency for the $\epsilon_\gamma \lesssim 50\%$ Subsample after GRB 110213A

$\epsilon_\gamma \lesssim 50\%$ Subsample (62 Bursts)									
GRB	z	$E_{p,z}$ (keV)	E_{iso} (10^{52} erg)	$L_{X,11}$ (10^{45} erg s $^{-1}$)	$E_{\text{iso}} / \left[\frac{3.7 * L_{X,11}}{(1+z)/2} \right]$	E_{ratio}	$(R_{\text{ph}}/R_s)^{-2/3}$	Γ	$T_{90,i}$ (s)
110213A	1.46	183.83 $\pm$ 32.15	6.9	3.22	0.713	0.225	0.283	157	11.9
110213B	1.083	256 $\pm$ 40	7.04	3.66	0.541	0.348			
110715A	0.82	216.58 $\pm$ 12.74	4.97	9.99	0.122	0.313			
111209A	0.677	520 $\pm$ 89	5.2	2.95	0.399	0.991			
111228A	0.7156	74 $\pm$ 53	1.65	3.24	0.118	0.056			
120326A	1.798	115 $\pm$ 19	3.82	15.6	0.093	0.147			
120804A	1.3	283 $\pm$ 62	0.657	1.09	0.187	0.877			
120811C	2.671	203.98 $\pm$ 19.55	8.81	7.25	0.603	0.239			
120907A	0.97	241.16 $\pm$ 67.27	0.18	1.02	0.047	1.09			
121211A	1.023	202.76 $\pm$ 32.05	0.14	2.54	0.015	0.942			
130420A	1.297	131.59 $\pm$ 7.2	6.29	2.99	0.653	0.149	0.644	149	45.7
130612A	2.006	186 $\pm$ 32	0.719	0.32	0.913	0.487	0.653	193	1.9
130701A	1.16	192.24 $\pm$ 8.64	2.62	1.32	0.580	0.33			
130702A	0.145	17.2 $\pm$ 5.7	0.064	1.01	0.010	0.028			
130831A	0.48	81.4 $\pm$ 13.32	0.805	1.02	0.157	0.156	0.154	73	11.9
130925A	0.347	110.94 $\pm$ 3.1	15.	14.6	0.187	0.089			
131117A	4.042	222 $\pm$ 37	1.03	1.43	0.491	0.546			
131231A	0.6439	292.52 $\pm$ 6.06	21.1	13.	0.361	0.288	0.142	145	17.7
140213A	1.2076	191.24 $\pm$ 7.85	8.88	13.4	0.197	0.218			
140226A	1.98	1234 $\pm$ 235	5.68	1.71	1.340	3.05			
140304A	5.283	775.06 $\pm$ 173.37	10.3	5.19	1.690	1.34			
140506A	0.889	373.19 $\pm$ 61.49	1.4	5.88	0.060	0.986			
140508A	1.027	533.46 $\pm$ 28.44	22.5	14.4	0.428	0.629			
140512A	0.72	826 $\pm$ 201.24	7.25	6.98	0.241	2.64			
140515A	6.32	376 $\pm$ 108	5.38	3.15	1.690	0.636			
140606B	0.384	352 $\pm$ 46	0.25	1.	0.047	1.62			
140620A	2.04	230.19 $\pm$ 33.87	7.28	11.4	0.263	0.299			
140801A	1.32	250.6 $\pm$ 7.	5.55	1.87	0.930	0.366			
140907A	1.21	313 $\pm$ 21	2.71	3.58	0.226	0.626			
141109A	2.993	763 $\pm$ 303	33.1	10.	1.780	0.891			
141221A	1.452	450.62 $\pm$ 87.15	2.46	1.11	0.734	1.05	0.965	202	9.7
141225A	0.915	342.71 $\pm$ 52.13	0.859	0.494	0.450	1.04			
150206A	2.09	704.52 $\pm$ 71.07	61.9	21.1	1.220	0.651			
150301B	1.5169	460.51 $\pm$ 90.95	1.99	0.567	1.190	1.16			
150323A	0.59	151.05 $\pm$ 14.31	1.26	0.427	0.632	0.306			
150403A	2.06	1311.74 $\pm$ 53.09	116.	43.9	1.100	1.21			
150514A	0.807	116.72 $\pm$ 10.19	0.878	1.26	0.171	0.245			
150727A	0.313	195.05 $\pm$ 25.18	0.2	0.062	0.575	0.794			
150818A	0.282	128 $\pm$ 37	0.1	0.09	0.193	0.571			
151027A	0.81	366.76 $\pm$ 61.78	3.3	6.17	0.131	0.724			
151029A	1.423	82 $\pm$ 17	0.288	0.232	0.407	0.221			
160227A	2.38	222 $\pm$ 55	5.56	14.6	0.174	0.311			
160509A	1.17	770.74 $\pm$ 20.82	113.	37.8	0.877	0.6			
160623A	0.37	756.24 $\pm$ 19.18	25.3	14.8	0.317	0.963			
160804A	0.736	123.93 $\pm$ 7.25	2.7	1.16	0.543	0.182			
161017A	2.013	718.83 $\pm$ 122.83	8.3	4.35	0.777	1.31			
161117A	1.549	205.62 $\pm$ 7.76	13.	7.04	0.637	0.212			
161219B	0.1475	71. $\pm$ 19.3	0.012	0.778	0.002	0.533			
170113A	1.968	333.92 $\pm$ 174.49	0.924	6.	0.061	0.976			
170604A	1.329	512 $\pm$ 168	4.7	4.61	0.321	1.			
170607A	0.557	174.06 $\pm$ 14.06	0.915	3.1	0.062	0.411			
170705A	2.01	294.61 $\pm$ 23.01	18.	25.3	0.289	0.307			
170903A	0.886	180 $\pm$ 25	0.865	3.02	0.073	0.438			
171205A	0.0368	125 $^{+141}_{-37}$	0.002	0.003	0.116	1.98			
171222A	2.409	694 $\pm$ 12	8.94	5.83	0.706	1.22			
180205A	1.409	205 $\pm$ 34	0.972	1.63	0.194	0.501			
180620B	1.1175	372 $\pm$ 105	3.04	12.4	0.070	0.758			
180720B	0.654	1052 $\pm$ 26	34.	32.	0.237	1.36			
180728A	0.117	108 $\pm$ 8	0.233	1.11	0.032	0.343			
181201A	0.45	220 $\pm$ 9	10.	11.7	0.168	0.253			
190106A	1.859	489 $\pm$ 257	9.96	14.1	0.272	0.735			
190114C	0.4245	929.3 $\pm$ 9.4	27.	9.3	0.560	1.24			

Table 3
The Observed Quantities and Inferred Efficiency for the $\epsilon_\gamma \gtrsim 50\%$ Subsample after GRB 110213A

$\epsilon_\gamma \gtrsim 50\%$ Subsample (40 Bursts)								
GRB	z	$E_{p,z}$ (keV)	E_{iso} (10^{52} erg)	$L_{X,11}$ (10^{45} erg s $^{-1}$)	$E_{\text{iso}} / \left[\frac{3.7 * L_{X,11}}{(1+z)^2} \right]$	η/Γ	Γ	L_{iso} (10^{52} erg s $^{-1}$)
110422A	1.77	429.35 $\pm$ 8.31	74.700	5.450	5.140			
110503A	1.61	574.2 $\pm$ 31.32	21.300	4.390	1.710			
110731A	2.83	1223 $\pm$ 75.4	31.500	4.650	3.510	1.722	500	20.52
110918A	0.98	665.28 $\pm$ 79.2	271.000	39.100	1.850			
111008A	5.	624 $\pm$ 186	41.400	9.390	3.580			
120119A	1.728	499.91 $\pm$ 21.71	40.200	4.100	3.610			
120711A	1.405	2552 $\pm$ 91	204.000	39.800	1.660	7.587	258	14.37
120712A	4.1745	642 $\pm$ 134.5	15.200	2.310	4.600			
120909A	3.93	961.41 $\pm$ 125.42	69.000	7.570	6.070	5.060	288	14.88
121128A	2.2	244.19 $\pm$ 9.61	10.100	1.000	4.340	3.891	332	6.64
130408A	3.76	1289.96 $\pm$ 204.68	32.400	5.570	3.740			
130427A	0.34	1105.4 $\pm$ 7.3	89.000	17.000	0.947	1.050	471	11.86
130505A	2.27	1939.11 $\pm$ 85.02	438.000	29.400	6.590			
130514A	3.6	506 $\pm$ 193	49.500	5.730	5.370			
130518A	2.488	1388.34 $\pm$ 55.23	216.000	6.000	17.000			
130610A	2.092	840.81 $\pm$ 344.98	5.780	0.645	3.750	6.465	204	1.3
130907A	1.24	866.88 $\pm$ 35.84	385.000	23.900	4.870			
131030A	1.29	448.84 $\pm$ 13.74	32.700	7.060	1.430			
131105A	1.686	713.18 $\pm$ 46.18	15.300	3.340	1.660			
140206A	2.73	1780.44 $\pm$ 119.77	278.000	26.800	5.220			
140419A	3.96	1398.72 $\pm$ 188.48	228.000	10.800	14.200			
140423A	3.26	516.74 $\pm$ 64.73	43.800	4.720	5.350	6.019	303	6.05
140629A	2.275	282 $\pm$ 56	4.400	0.575	3.390			
140703A	3.14	861.25 $\pm$ 148.3	18.400	3.280	3.130			
141004A	0.573	231 $\pm$ 44	0.210	0.051	0.877			
141028A	2.33	979.59 $\pm$ 53.39	51.000	7.460	3.070			
141220A	1.3195	418.8 $\pm$ 24.17	2.290	0.322	2.230			
150314A	1.758	957.28 $\pm$ 19.06	76.800	5.090	5.620			
151021A	2.33	566.1 $\pm$ 43.29	113.000	4.470	11.400			
160131A	0.97	1282.47 $\pm$ 453.1	87.000	6.990	3.310			
160625B	1.406	1134.34 $\pm$ 15.51	510.000	39.200	4.230			
161023A	2.708	604.4 $\pm$ 137.2	68.000	5.900	5.780			
161129A	0.645	240.84 $\pm$ 70.09	0.783	0.059	2.970			
170202A	3.645	1147 $\pm$ 771	17.000	4.420	2.410			
171010A	0.3285	227.17 $\pm$ 9.3	18.000	2.880	1.123			
180325A	2.248	993.89 $\pm$ 162.4	23.000	1.300	7.770			
180329B	1.998	146 $\pm$ 28	4.690	0.995	1.910			
180914B	1.096	977 $\pm$ 61	370.000	7.600	13.800			
181020A	2.938	1461 $\pm$ 225	82.800	4.560	9.660			
181110A	1.505	120 $\pm$ 68	11.000	0.937	3.970			

Also, the bursts with an early multi-peaked optical light curve or an optical peak preceded by a decreasing light curve should be excluded (see Ghirlanda et al. 2018).

2.2. Scaling Relations Predicted by the Photosphere Model

For the photosphere emission model, $\epsilon_\gamma \gtrsim 50\%$ and $\epsilon_\gamma \lesssim 50\%$ should correspond to the unsaturated acceleration case ($\Gamma \lesssim \eta$; $R_{\text{ph}} < R_s$; $E_{\text{iso}}/E_k = \eta Mc^2/\Gamma Mc^2 = \eta/\Gamma \gtrsim 1$) and the saturated acceleration case ($\Gamma = \eta$; $R_{\text{ph}} > R_s$; $E_{\text{iso}}/E_k = [\eta Mc^2(R_{\text{ph}}/R_s)^{-2/3}]/\Gamma Mc^2 = (R_{\text{ph}}/R_s)^{-2/3} \lesssim 1$), respectively.

2.2.1. $E_{\text{iso}}/E_k = \eta/\Gamma$ for the Unsaturated Acceleration Case, Corresponding to $\epsilon_\gamma \gtrsim 50\%$

When the outflow Lorentz factor Γ at the photosphere radius R_{ph} is less than the baryon loading η ($\eta = E/Mc^2$, where E and M are the injected energy and the baryon mass at the outflow base, respectively), the photosphere emission is in the

unsaturated acceleration case (or $R_{\text{ph}} < R_s$; where $R_s = \eta r_0$ and r_0 is the initial acceleration radius). In this case, $\Gamma = R_{\text{ph}}/r_0$.

For the unsaturated acceleration case, the observed temperature $T_{\text{ob}} = D \cdot T_{\text{comoving}} = \Gamma \cdot (T_0/\Gamma) = T_0$. Here, D is the Doppler factor, T_{comoving} is the photon temperature in the outflow comoving frame, and T_0 is the temperature at the outflow base r_0 . Thus, $E_{\text{iso}} = E = \eta Mc^2$. Also, because all the thermal energy is released at the photosphere radius (where there is no remaining energy to accelerate the jet), the Lorentz factor in the afterglow phase will remain as Γ , namely $E_k = \Gamma Mc^2$. So we should have $E_{\text{iso}}/E_k = \eta Mc^2/\Gamma Mc^2 = \eta/\Gamma \gtrsim 1$ (see Figures 1(b) and 2(e)), corresponding to $\epsilon_\gamma \gtrsim 50\%$.

2.2.2. $E_p \propto (E_{\text{iso}})^{1/4}$ for $\epsilon_\gamma \gtrsim 50\%$

For the photosphere emission model, the peak energy of the observed spectrum E_p corresponds to the temperature of the observed blackbody T_{ob} . In the unsaturated regime ($R_{\text{ph}} < R_s$,

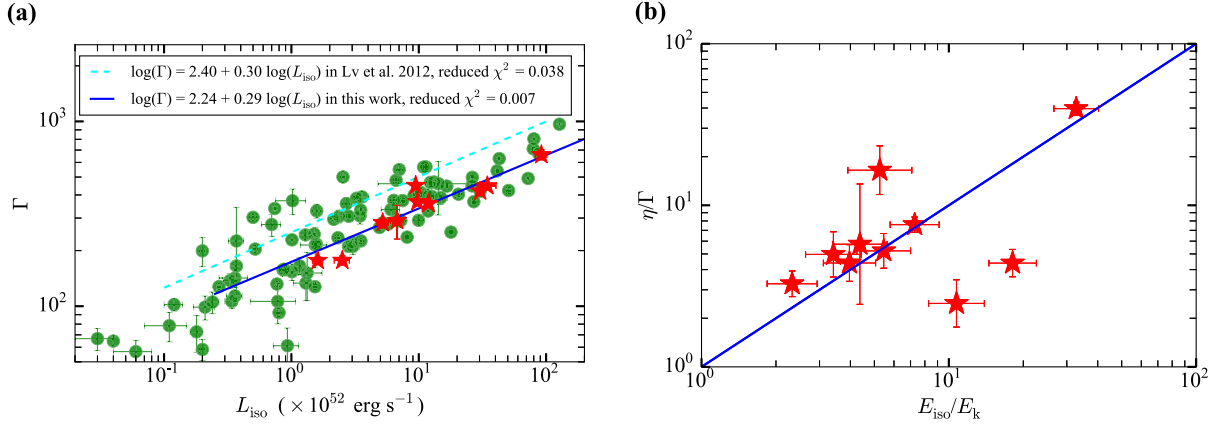


Figure 1. The $L_{\text{iso}} - \Gamma$ correlation and the $E_{\text{iso}}/E_k - \eta/\Gamma$ correlation for the selected extremely high-efficiency GRBs ($\epsilon_\gamma \gtrsim 80\%$). (a) The correlation of L_{iso} and Γ . Obviously, the tight correlation of $\Gamma \propto (L_{\text{iso}})^{0.29}$ (red stars; reduced $\chi^2 = 0.007$) is found, which is consistent with the prediction of the neutrino annihilation from the hyperaccretion disk, $\Gamma \propto (L_{\text{iso}})^{0.27} = (L_{\text{iso}})^{0.26}$. Thus, the jet is likely to be thermal-dominated. The dashed cyan line shows the $L_{\text{iso}} - \Gamma$ correlation found in Lü et al. (2012; reduced $\chi^2 = 0.038$, for the large Γ sample of green circles; Xue et al. 2019). (b) The correlation of E_{iso}/E_k and η/Γ . A significant linear correlation is found, and it is almost the same when we take $E_{k,52} = 5^* L_{X,45}$ (reduced $\chi^2 = 0.138$). This is consistent with the $E_{\text{iso}}/E_k = \eta Mc^2 / \Gamma Mc^2 = \eta / \Gamma$ predicted by the photosphere emission model in the unsaturated acceleration regime.

$\epsilon_\gamma \gtrsim 50\%$), $T_{\text{ob}} = T_0$. Since $T_0 \propto E^{1/4} \propto (E_{\text{iso}})^{1/4}$, we should have $E_p \propto (E_{\text{iso}})^{1/4}$ (see Figures 3–5; for $\epsilon_\gamma \gtrsim 80\%$ and $\epsilon_\gamma \gtrsim 50\%$ of long GRBs, as well as short GRBs).

$$\begin{aligned} 2.2.3. \text{ For } \epsilon_\gamma \lesssim 50\%, E_{\text{iso}} &= E \cdot (R_{\text{ph}}/R_s)^{-2/3}, \\ \text{and } E_p &\propto E^{1/4} \cdot (R_{\text{ph}}/R_s)^{-2/3} \end{aligned}$$

For $\epsilon_\gamma \lesssim 50\%$ ($R_{\text{ph}} > R_s$), the outflow performs adiabatic expansion at $r > R_s$. Thus, the comoving temperature decreases as $T_{\text{comoving}} = (T_0/\Gamma) \cdot (r/R_s)^{-2/3}$. The escaped photons at R_{ph} have $E_{\text{iso}} = E \cdot (R_{\text{ph}}/R_s)^{-2/3}$ and $E_p = 2.7kT_0 \cdot (R_{\text{ph}}/R_s)^{-2/3} \propto E^{1/4} \cdot (R_{\text{ph}}/R_s)^{-2/3}$. So E_{iso} and E_p should both decrease by the same factor of $(R_{\text{ph}}/R_s)^{-2/3}$, compared with the $E_p \propto (E_{\text{iso}})^{1/4}$ correlation for $\epsilon_\gamma \gtrsim 50\%$ (see Figures 4 and 5).

$$\begin{aligned} 2.2.4. \text{ For } \epsilon_\gamma \lesssim 50\%, \\ E_{\text{iso}}/E_k &= (R_{\text{ph}}/R_s)^{-2/3} = [(E_p/2.7k)^4 * (4\pi r_1^2 ac)/E_{\text{iso}}]^{1/3} \end{aligned}$$

For the $\epsilon_\gamma \lesssim 50\%$ sample (saturated acceleration, $E_k = E$), we should have

$$E_{\text{iso}} = E \cdot (R_{\text{ph}}/R_s)^{-2/3} \quad (1)$$

and

$$\begin{aligned} E_p &= 2.7kT_0 \cdot (R_{\text{ph}}/R_s)^{-2/3} \\ &= 2.7k(L/4\pi r_0^2 ac)^{1/4} \cdot (R_{\text{ph}}/R_s)^{-2/3} \\ &= 2.7k(E/4\pi r_1^2 ac)^{1/4} \cdot (R_{\text{ph}}/R_s)^{-2/3} \\ &= 2.7k(4\pi r_1^2 ac)^{-1/4} \cdot E^{1/4} \cdot (R_{\text{ph}}/R_s)^{-2/3}, \end{aligned} \quad (2)$$

where $r_1^2 \simeq r_0^2 * T_{90}$. From the relation of $\log(E_p) = 2.54 + 0.25 \log(E_{\text{iso}})$ in Figure 3(b), we obtain $r_1 = 8.45 \times 10^8$ cm. Then, from Equation (1) and Equation (2), we get

$$(E_p)^4/E_{\text{iso}} = (2.7k)^4 * (4\pi r_1^2 ac)^{-1} \cdot (R_{\text{ph}}/R_s)^{-2}, \quad (3)$$

thus

$$(R_{\text{ph}}/R_s)^{-2/3} = [(E_p/2.7k)^4 * (4\pi r_1^2 ac)/E_{\text{iso}}]^{1/3}. \quad (4)$$

Combined with the above mentioned $E_{\text{iso}}/E_k = (R_{\text{ph}}/R_s)^{-2/3}$, $E_{\text{ratio}} = E_{\text{iso}}/E_k = (R_{\text{ph}}/R_s)^{-2/3}$ is predicted ($E_{\text{ratio}} = [(E_p/2.7k)^4 * (4\pi r_1^2 ac)/E_{\text{iso}}]^{1/3}$; see Figures 2, and 5–7).

$$2.2.5. \Gamma \propto E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4} \text{ Correlation}$$

As in the previous statement—and shown in Figure 7(b)—for $\epsilon_\gamma \lesssim 50\%$, we have $E_{\text{ratio}} = (R_{\text{ph}}/R_s)^{-2/3}$. Considering

$$R_{\text{ph}} = \frac{\sigma_T}{8\pi m_p c^3} \frac{L}{\Gamma^3} \quad (5)$$

and $R_s = \Gamma \cdot r_0$, we have

$$[(E_p/2.7k)^4 * (4\pi r_1^2 ac)/E_{\text{iso}}]^{1/3} = \left(\frac{\sigma_T}{8\pi m_p c^3} \frac{E_{\text{iso}}/T_{90}}{\Gamma^4 \cdot r_0} \right)^{-2/3}. \quad (6)$$

Thus, we can use the quantities of the prompt emission (E_{iso} , E_p , and T_{90}) to estimate the Lorentz factor Γ , just as the obtained $\Gamma = 10^{3.33} L_{\text{iso}}^{0.46} E_p^{-0.43}$ is estimated from the statistic fitting in Liang et al. (2015) (see Figures 7(c) and 8). Based on Equation (6), we derive

$$\Gamma \propto E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4}. \quad (7)$$

Considering the constants, we obtain

$$\Gamma = 19.67 \cdot E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4}. \quad (8)$$

For the $\epsilon_\gamma \gtrsim 50\%$ case, from

$$R_{\text{ph}} = \left[\frac{\sigma_T}{6m_p c} \frac{L_{\text{iso}}}{4\pi c^2 \eta} r_0^2 \right]^{1/3}, \quad (9)$$

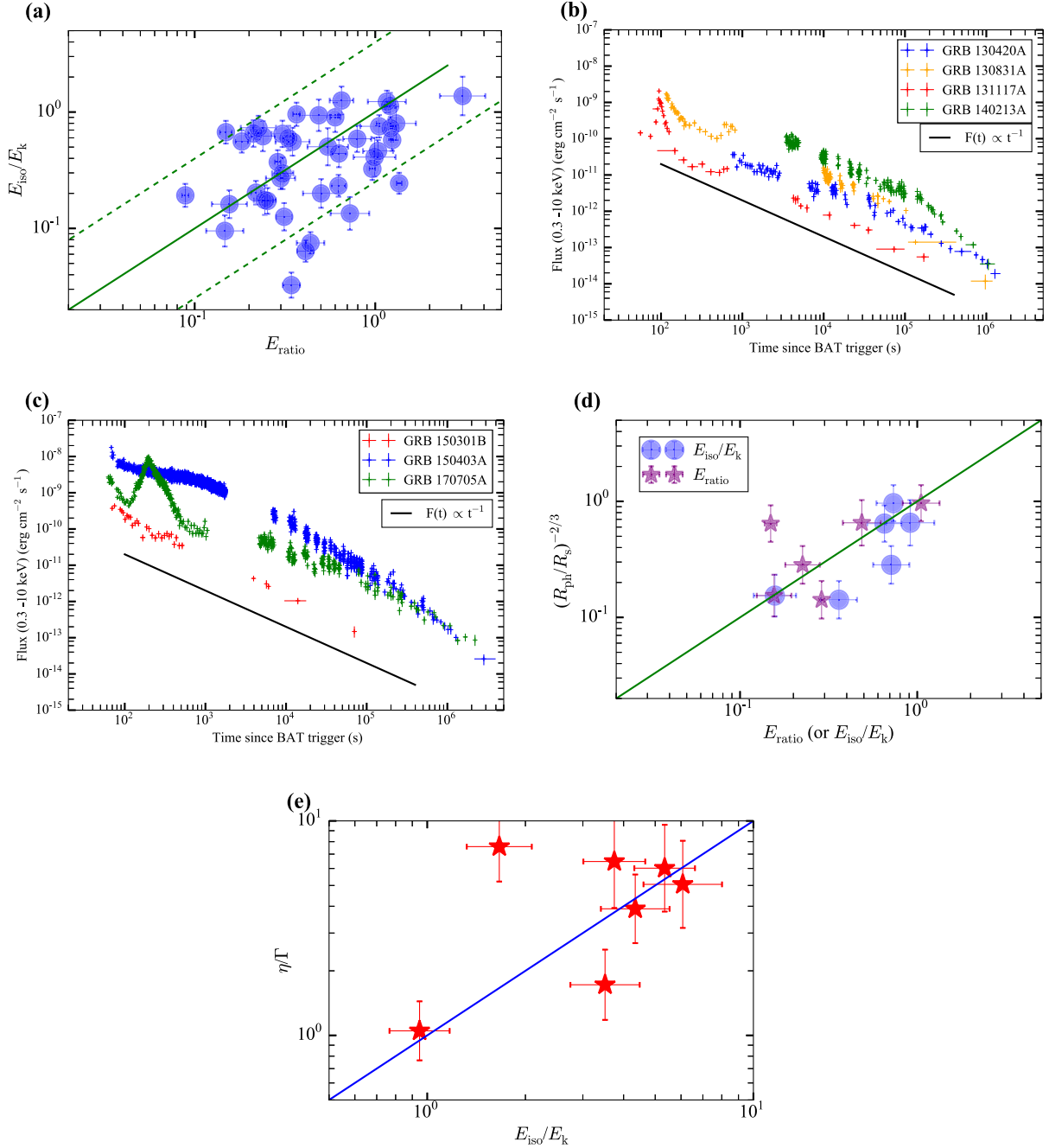


Figure 2. The similar observed efficiency (with E_{iso}/E_k from the X-ray afterglow) and theoretically predicted efficiency (from the prompt emission or the optical afterglow) of the photosphere emission model for the bursts after GRB 110213A. (a) The distribution of E_{ratio} ($E_{\text{ratio}} = [(E_p/2.7k)^4 * (4\pi r_1^2 ac)/E_{\text{iso}}]^{1/3}$, from the prompt emission) and E_{iso}/E_k for the $\epsilon_\gamma \lesssim 50\%$ sample (see Table 2). They are found to be centered around the equal-value line, and have a linear correlation (reduced $\chi^2 = 0.172$). The dispersion is likely to be caused by the estimation errors for E_k , since many X-ray afterglow light curves are not power laws with slopes of -1 . (b) and (c) The X-ray afterglow light curves for the bursts (seven bursts) with almost the same E_{ratio} and E_{iso}/E_k (see Table 5). We find that all of these light curves do have a power-law shape with a slope of ~ -1 . (d) The distributions of E_{ratio} , E_{iso}/E_k , and $(R_{\text{ph}}/R_s)^{-2/3}$ for the sample (six bursts; see Table 2), with detections of the peak time of the optical afterglow (to estimate Γ and thus $(R_{\text{ph}}/R_s)^{-2/3}$). It is found that one burst has almost the same values for these three quantities, and that the five other bursts have almost the same values for two quantities (reduced $\chi^2 = 0.089$ for all 12 markers). (e) The consistent efficiency from the observations (with E_{iso}/E_k) and the prediction from the photosphere emission model (with η/Γ) for the $\epsilon_\gamma \gtrsim 50\%$ sample, with detections of the peak time of the optical afterglow (seven bursts; see Table 3; reduced $\chi^2 = 0.120$).

we have

$$\begin{aligned} \Gamma = R_{\text{ph}}/r_0 &= \left[\frac{\sigma_T}{6m_p c} \frac{L_{\text{iso}}}{4\pi c^2 \eta} r_0^2 \right]^{1/3} / r_0 \\ &= \left[\frac{\sigma_T}{6m_p c} \frac{L_{\text{iso}}}{4\pi c^2 \eta r_0} \right]^{1/3}. \end{aligned} \quad (10)$$

Thus, $L_{\text{iso}} \propto E_{\text{iso}}/(T_{90}) \propto \eta \cdot \Gamma^3$. Combined with $E_{\text{iso}}/E_k = \eta M c^2 / \Gamma M c^2 = \eta / \Gamma$, we have $1/T_{90} \propto (1/E_k) \cdot \Gamma^4$, namely $E_k \propto \Gamma^4 \cdot T_{90}$. Putting this into $\Gamma \propto (E_k)^{1/8} \cdot [T_p/(1+z)]^{-3/8}$, we obtain $\Gamma \propto \Gamma^{1/2} \cdot (T_{90})^{1/8} \cdot [T_p/(1+z)]^{-3/8}$, namely $\Gamma^{1/2} \cdot (T_{90})^{-1/8} \propto [T_p/(1+z)]^{-3/8}$. From Figure 9(a) (discussed in Section 2.2.7), we have $\Gamma \propto E_p/(T_{90})^{1/4}$. So, as in the $\epsilon_\gamma \lesssim 50\%$ case, $E_p^{1/2}/(T_{90})^{1/4} \propto [T_p/(1+z)]^{-3/8}$ is obtained

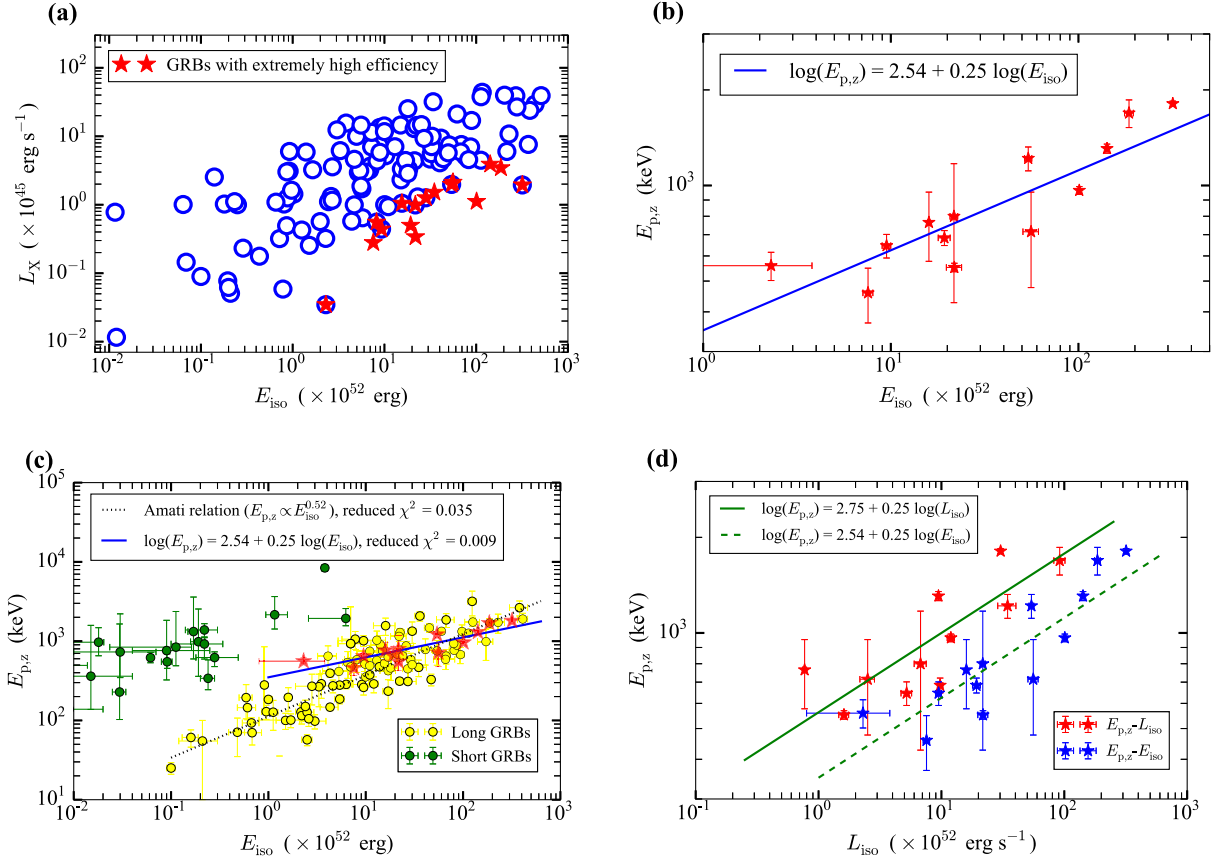


Figure 3. The $E_p \propto (E_{\text{iso}})^{1/4}$ and $E_p \propto (L_{\text{iso}})^{1/4}$ relations for the GRBs with the highest prompt efficiency ϵ_γ ($\epsilon_\gamma \gtrsim 80\%$; see Table 1). (a) The E_{iso} and calculated $L_{X,11\text{h}}$ (the late-time X-ray afterglow luminosity at 11 hr) distribution for the whole GRB sample used (117 bursts, after GRB 110213A) with redshift. The red stars with the smallest $L_{X,11\text{h}}$ represent the bursts with the highest prompt efficiency. (b) The E_p and E_{iso} distribution for the 15 long GRBs selected. The best-fit result is $\log(E_p) = 2.54 + 0.25 \log(E_{\text{iso}})$, which is quite consistent with the $E_p \propto (E_{\text{iso}})^{1/4}$ relation predicted by the photosphere (thermal) emission model. (c) Comparison of the E_p and E_{iso} distributions for the selected GRBs (red stars) and the large sample of long GRBs (yellow circles; see Figure 3 in Zhang et al. 2018b). Obviously, the dispersion of the selected GRBs is quite small (reduced $\chi^2 = 0.009$) relative to that of the large sample (reduced $\chi^2 = 0.035$). The black dotted line shows the Amati relation (Amati et al. 2002) for the large sample. (d) Comparison of the $E_p - L_{\text{iso}}$ distribution (red stars) and the $E_p - E_{\text{iso}}$ distribution (blue stars) for the selected GRBs. Likewise, the $E_p \propto (L_{\text{iso}})^{1/4}$ relation exists. The best-fit result is $\log(E_p) = 2.75 + 0.23 \log(L_{\text{iso}})$. The dispersion (reduced $\chi^2 = 0.012$) is found to be similar to that of $E_p \propto (E_{\text{iso}})^{1/4}$.

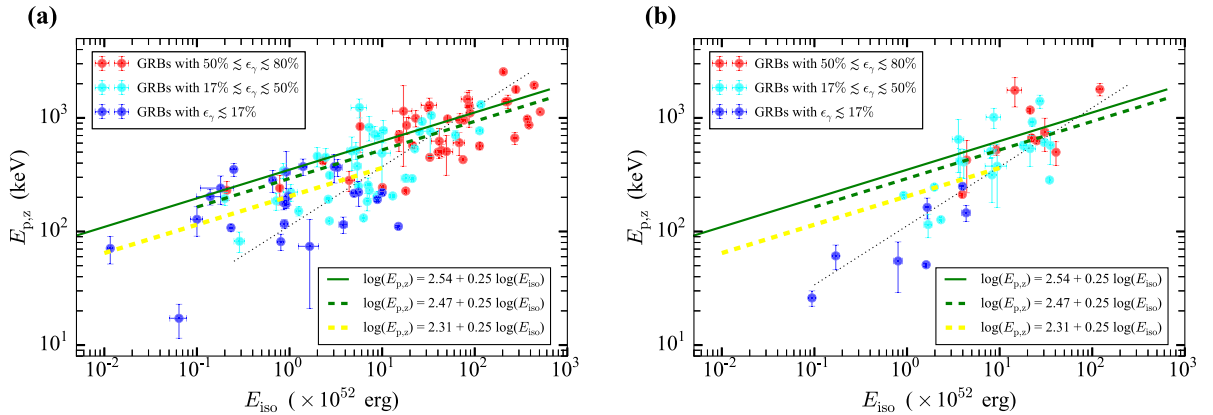


Figure 4. The different distributions of E_p and E_{iso} for the two distinguished samples ($\epsilon_\gamma \gtrsim 50\%$ and $\epsilon_\gamma \lesssim 50\%$; see Tables 2 and 3) for the long GRBs. (a) The E_p and E_{iso} distribution for the bursts after GRB 110213A (117 bursts, with $L_{X,11\text{h}}$ derived in this work). For the $\epsilon_\gamma \gtrsim 50\%$ sample (red circles), the best-fit result is $\log(E_p) = 2.47 + 0.25 \log(E_{\text{iso}})$ (reduced $\chi^2 = 0.039$; for the dotted black line of the typical Amati relation, reduced $\chi^2 = 0.079$), which is consistent with the predicted $E_p \propto (E_{\text{iso}})^{1/4}$ from the photosphere emission model. For the $\epsilon_\gamma \lesssim 50\%$ sample (blue and cyan circles), the upmost distribution is found around $\log(E_p) = 2.54 + 0.25 \log(E_{\text{iso}})$, and the best-fit result is $\log(E_p) = 2.31 + 0.26 \log(E_{\text{iso}})$ (reduced $\chi^2 = 0.076$; for the typical Amati relation, reduced $\chi^2 = 0.113$), much below that. (b) The E_p and E_{iso} distribution for the bursts before GRB 110213A (46 bursts, with $L_{X,11\text{h}}$ given in D’Avanzo et al. 2012). Different distributions of E_p and E_{iso} for the two distinguished samples ($\epsilon_\gamma \gtrsim 50\%$ and $\epsilon_\gamma \lesssim 50\%$) are also found, similar to the distributions for the bursts after GRB 110213A.

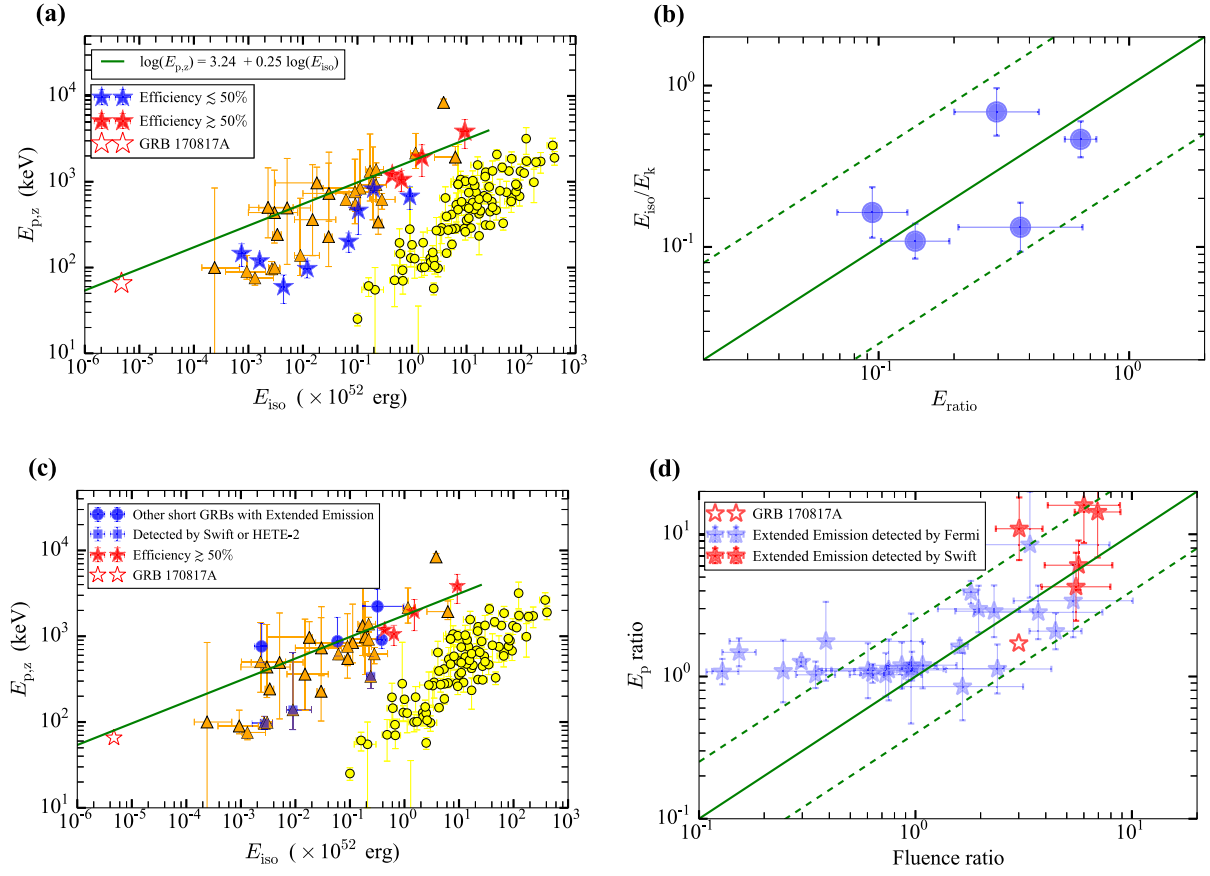


Figure 5. Evidence from the short GRBs (see Table 8). (a) The different distributions of E_p and E_{iso} for the two distinguished samples: $\epsilon_\gamma \gtrsim 50\%$ (red stars) and $\epsilon_\gamma \lesssim 50\%$ (blue stars). (b) The distribution of E_{ratio} and E_{iso}/E_k for the $\epsilon_\gamma \lesssim 50\%$ sample (reduced $\chi^2 = 0.140$). (c) The E_p and E_{iso} distributions of the main pulse for seven other bursts (blue circles and boxes) that have extended emission (see Minaev & Pozanenko 2020) and lack efficiency. Note that the five bursts of the $\epsilon_\gamma \gtrsim 50\%$ sample (red stars) all have extended emission, including GRB 170817A. (d) A comparison of the ratios of E_p and the fluence (or E_{iso}) for the main pulse and the extended emission of a large extended emission sample (reduced $\chi^2 = 0.196$), including the bursts without redshift (blue stars).

(as discussed in Section 2.2.6). This means that Equation (7) is also available for the $\epsilon_\gamma \gtrsim 50\%$ case. After considering the constants, we get

$$\Gamma = 16.75 \cdot E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4}. \quad (11)$$

Note that different constants for the $\epsilon_\gamma \lesssim 50\%$ case and the $\epsilon_\gamma \gtrsim 50\%$ case are predicted (see Figures 7(d) and (e)). Also, the calculation of Γ in Figure 9(a), using the optical afterglow, has assumed an efficiency of $\sim 20\%$ (namely, $E_k = 5 \cdot E_{\text{iso}}$). In fact, when calculating Γ , we should modify E_k , based on the real efficiency (η/Γ). For the $\epsilon_\gamma \gtrsim 50\%$ case, we typically have $\eta/\Gamma \sim 3.5$ (see Figures 9(a) and 10). Thus, the constant ~ 16.75 should be ~ 1.4 times smaller.

2.2.6. $T_p/(1+z) \propto [E_p/(T_{90})^{1/2}]^{-4/3}$ Correlation

Note that when using the optical afterglow (the peak time T_p) to obtain Γ , we have

$$\begin{aligned} \Gamma &\propto (E_k)^{1/8} \cdot [T_p/(1+z)]^{-3/8} \\ &\propto (E_{\text{iso}}/\epsilon_\gamma)^{1/8} \cdot [T_p/(1+z)]^{-3/8}, \end{aligned} \quad (12)$$

where ϵ_γ is normally taken as 0.2. Then, if Equation (7) is correct, combining Equation (7) and Equation (12) should give

(see Figure 7(d))

$$T_p/(1+z) \propto [E_p/(T_{90})^{1/2}]^{-4/3}. \quad (13)$$

2.2.7. $\Gamma \propto E_p/(T_{90})^{1/4}$ Correlation

As shown in Figure 4(a), we have $E_p \propto (E_{\text{iso}})^{1/4}$ for both the $\epsilon_\gamma \gtrsim 50\%$ case (smaller dispersion) and the $\epsilon_\gamma \lesssim 50\%$ case (larger dispersion). Then, with Equation (7), we should obtain $\Gamma \propto E_p/(T_{90})^{1/4}$. On the other hand, from Figure 1(a) and Figure 11(a), we have approximately $\Gamma \propto (L_{\text{iso}})^{1/4}$, thus $\Gamma \propto E_p/(T_{90})^{1/4}$ is also likely to be obtained (see Figure 9).

2.3. Tightness of the Scaling Relations and Data Errors

To quantitatively measure the tightness of the scaling relations, we calculate the statistical value

$$\chi^2 = \sum_{i=1}^N (\lg y_i - a \lg x_i - b)^2 \quad (14)$$

for each relation of $\lg y = a \lg x + b$ (for example, $\lg E_p = 0.25 \lg E_{\text{iso}} + 2.54$). The reduced $\chi^2 = \chi^2/\text{degrees of freedom}$ (dof = $N - 2$) for each relation is given in the caption for the corresponding figure. Note that $\sqrt{\chi^2/\text{dof}}$ is approximate to the typical dispersion measure σ (both in units of dex). In

Table 4
The Fitting Results of the Time-integrated Spectra for Bursts with Extremely High Efficiency (Detected by Fermi)

GRB	α (CPL)	E_c (CPL)	F (CPL)	α (Band)	β (Band)	E_p (Band)	F (Band)	Δ DIC (Band-CPL)	P_{DIC} (CPL)	P_{DIC} (Band)
110818A	$-1.18^{+0.15}_{-0.15}$	352^{+169}_{-161}	$0.18^{+0.12}_{-0.06} \times 10^{-6}$	$0.48^{+0.50}_{-0.56}$	$-1.70^{+0.07}_{-0.07}$	42^{+7}_{-7}	$0.46^{+1.54}_{-0.35} \times 10^{-6}$	-141.4	-2.9	-151.3
120729A	$-0.55^{+0.04}_{-0.04}$	163^{+23}_{-24}	$0.17^{+0.05}_{-0.04} \times 10^{-6}$	$-0.55^{+0.04}_{-0.04}$	$-4.93^{+2.44}_{-2.66}$	229^{+30}_{-29}	$0.19^{+0.07}_{-0.04} \times 10^{-6}$	-1.3	1.9	1.4
131108A	$-0.93^{+0.02}_{-0.02}$	380^{+25}_{-26}	$1.36^{+0.12}_{-0.10} \times 10^{-6}$	$0.66^{+0.08}_{-0.08}$	$-1.60^{+0.01}_{-0.01}$	61^{+0}_{-0}	$3.54^{+0.69}_{-0.61} \times 10^{-6}$	599.9	2.9	1.7
150821A	$-1.19^{+0.02}_{-0.02}$	419^{+35}_{-35}	$0.64^{+0.05}_{-0.05} \times 10^{-6}$	$-1.18^{+0.03}_{-0.03}$	$-3.87^{+1.71}_{-3.16}$	313^{+31}_{-32}	$0.76^{+0.19}_{-0.15} \times 10^{-6}$	-8.0	2.9	-1.5
161014A	$-0.74^{+0.10}_{-0.10}$	144^{+24}_{-24}	$0.22^{+0.07}_{-0.05} \times 10^{-6}$	$1.52^{+0.34}_{-0.34}$	$-1.74^{+0.04}_{-0.04}$	48^{+2}_{-2}	$0.63^{+0.82}_{-0.37} \times 10^{-6}$	3.0	1.6	-40.0
170214A	$-1.06^{+0.01}_{-0.01}$	570^{+26}_{-26}	$1.53^{+0.08}_{-0.06} \times 10^{-6}$	$-1.05^{+0.01}_{-0.01}$	$-2.77^{+0.36}_{-0.12}$	510^{+22}_{-22}	$1.81^{+0.15}_{-0.14} \times 10^{-6}$	-10.6	3.0	2.7

Note. For 120729A, 150821A, and 170214A, the high-energy spectral indices for the BAND function are very small (-4.93 , -3.87 , and -2.77), while the low-energy spectral indices and peak energy for the BAND function and CPL model are very similar. For 110818A, 131108A, and 161014A, the low-energy spectral indices for the BAND function are extremely hard (0.48, 0.66, and 1.52), and the peak energies are extremely small (42, 61, and 48). Thus, the CPL model is surely the best-fit model.

Table 5
The Bursts with Almost the Same E_{ratio} and E_{iso}/E_k ($\epsilon_\gamma \lesssim 50\%$), and the Power-law Slopes of Their X-Ray Afterglow Light Curves

Bursts after GRB 110213A				Bursts before GRB 110213A			
GRB	$E_{iso} / \left[\frac{3.7 * L_{X,11}}{(1+z)/2} \right]$	E_{ratio}	X-Ray Slope	GRB	$E_{iso} / \left[\frac{3.7 * L_{X,11}}{(1+z)/2} \right]$	E_{ratio}	X-Ray Slope
130420A	0.653	0.644 ^a	-0.900	050922C	0.898	0.821	-1.200
130831A	0.157	0.156	-0.959	060210	0.751	0.597	-0.970
131117A	0.491	0.546	-0.998	060306	0.567	0.435	-1.047
140213A	0.197	0.218	-1.070	080603B	0.629	0.528	-0.850
150301B	1.190	1.16	-1.179	091018	0.112	0.092	-1.160
150403A	1.100	1.21	-1.140	091020	0.912	0.831	-1.090
170705A	0.289	0.307	-0.969	100621A	0.159	0.193	-0.987

Note.

^a $E_{iso}/E_k = (R_{ph}/R_s)^{-2/3}$.

Figure 10, for the Gaussian fit, $\chi^2 = \sum (y_i - \bar{y}_{i,model})^2$ is adopted.

To show how well the data follows the predicted relations, the missing errors for the considered quantities are estimated. For the compound quantities (such as E_{iso}/E_k), the errors are estimated by error propagation: $\Delta[\lg(E_{iso}/E_k)] = \Delta[\lg(E_{iso}) - \lg(E_k)] = \Delta[\lg(E_{iso})] + \Delta[\lg(E_k)]$. For the $L_{X,11h}$ derived in this work, $T_{p,op}$, and $T_{90,i}$, the errors at the 90% confidence level (~ 0.1 dex) are taken, just as $\Delta(L_{X,11h})$ given in D’Avanzo et al. (2012).

3. Evidence from Long GRBs with Extremely High Efficiency ($\epsilon_\gamma \gtrsim 80\%$)

There is much controversy about the spectral differences between the photosphere emission model and the synchrotron emission model, after considering the natural and complicated physical conditions (jet structure, decaying magnetic field, and so on; e.g., Uhm & Zhang 2014; Geng et al. 2018; Meng et al. 2018, 2019, 2022). Nevertheless, a crucial difference between these two models is that the photosphere emission model predicts much higher radiation efficiency ϵ_γ . The synchrotron emission models mainly include the internal shock model (for a matter-dominated fireball; Rees & Meszaros 1994) and the ICMART model (internal-collision-induced magnetic reconnection and turbulence, for a Poynting flux-dominated outflow; Zhang & Yan 2011). For the internal shock model, since only the relative kinetic energy between different shells can be released, the radiation efficiency is rather low ($\sim 10\%$; Kobayashi et al. 1997). For the ICMART model, the radiation

efficiency can be much higher ($\sim 50\%$), and it reaches $\sim 80\%$ in the extreme case. However, the extremely high ϵ_γ ($\epsilon_\gamma \gtrsim 80\%$) is unlikely to be achieved, because the magnetic reconnection requires some conditions to be triggered, so much magnetic energy is left. For the photosphere emission model, if only the acceleration is in the unsaturated regime ($R_{ph} < R_s$), the radiation efficiency can be close to 100%. Thus, in this work, we select GRBs with extremely high ϵ_γ ($\epsilon_\gamma \gtrsim 80\%$; see Figure 3(a) and Table 1). Note that the two bursts (GRB 990705 and GRB 000210) that are claimed to have extremely high ϵ_γ in Lloyd-Ronning & Zhang (2004) are also included. We then analyze the prompt⁴ and afterglow⁵ properties of these 15 long GRBs, to confirm the photosphere emission origin.

3.1. Characteristics of Prompt Emission

In Figure 3(b), we plot the E_p and E_{iso} distributions of the selected GRBs (omitting GRB 081203A and GRB 130606A, because of the large E_p error), and we find that they follow the predicted $E_p \propto (E_{iso})^{1/4}$ relation (see Section 2.2.2) quite well. The best-fit result is $\log(E_p) = 2.54 + 0.25 \log(E_{iso})$. In Figure 3(c), we compare the E_p and E_{iso} distributions of the

⁴ The Fermi/GBM data are publicly available at <https://heasarc.gsfc.nasa.gov/W3Browse/fermi/fermigbrst.html>. The Konus-Wind data are publicly available at <https://vizier.cds.unistra.fr/viz-bin/VizieR?source=J/ApJ/850/161>.

⁵ The X-ray afterglow data are publicly available at https://www.swift.ac.uk/xt_products/. The optical afterglow data are taken from Li et al. (2012, 2018) and Liang et al. (2013).

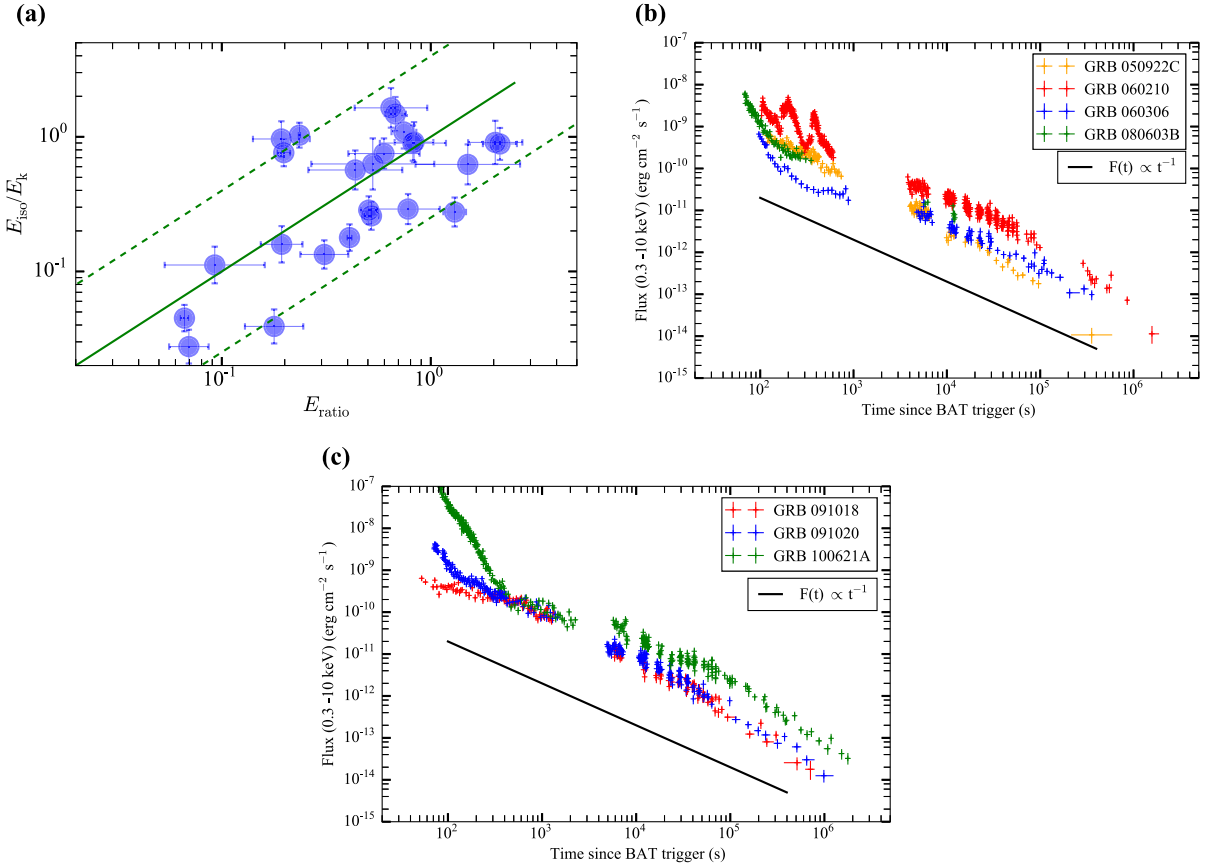


Figure 6. The similar observed efficiency (with E_{iso}/E_k from the X-ray afterglow) and theoretically predicted efficiency (from the prompt emission) for the bursts before GRB 110213A (with $L_{X,11h}$ as given in D’Avanzo et al. 2012). (a) The distribution of E_{ratio} and E_{iso}/E_k for the $\epsilon_\gamma \lesssim 50\%$ sample, which is found to be centered around the equal-value line and have a linear correlation (reduced $\chi^2 = 0.160$). (b), (c) The X-ray afterglow light curves for the bursts (seven bursts) with almost the same E_{ratio} and E_{iso}/E_k (see Table 5). We find that all these light curves have a power-law shape with a slope of ~ -1 .

selected GRBs with those of the large sample of long GRBs, and find that the dispersion is quite small relative to that of the large sample.

In Figure 3(d), we plot the E_p and L_{iso} distributions of the selected GRBs, and find that they also follow the $E_p \propto (L_{\text{iso}})^{1/4}$ relation well. The best-fit result is $\log(E_p) = 2.75 + 0.23 \log(L_{\text{iso}})$. And the dispersion is found to be similar to that of $E_p \propto (E_{\text{iso}})^{1/4}$. Furthermore, based on the best-fit $E_p \sim 10^{2.75} \cdot (L_{\text{iso}})^{1/4}$ and $E_p = 2.7kT_0 = 2.7k(L_{\text{iso}}/4\pi r_0^2 ac)^{1/4}$, we obtain the initial acceleration radius $r_0 \sim 3.21 \times 10^8$ cm, which is consistent with the quite high mean value $\langle r_0 \rangle \sim 10^{8.5}$ cm deduced in Pe’er et al. (2015).

In Figure 12, we show the E_p evolutions of the time-resolved spectra for five GRBs detected by Fermi/GBM. The E_p evolutions are found to follow the evolution of the flux F quite well (intensity-tracking pattern; Liang & Kargatis 1996). This positive correlation is consistent with the abovementioned unsaturated acceleration condition of the photosphere emission. From Tables 1 and 4, we can see that the best-fit spectral model of the time-integrated spectra is the CPL model, or that the high-energy spectral index β (using the BAND function to fit) is minimal. Thus, the photosphere emission model can better explain the high-energy spectra of these high-efficiency GRBs.

3.2. Characteristics of Afterglows

Figure 13 shows the X-ray afterglow light curves of the selected GRBs (excluding GRB 990705 and GRB 000210). We find that all the X-ray afterglow light curves appear as

simple power-law shapes,⁶ without any plateaus, steep decays (Zhang et al. 2006), or significant flares (with weak flares in the early times). In Figure 14, we show the optical afterglow light curves of six GRBs whose early peaks can be detected. All the optical afterglow light curves show significant reverse shock signals. The power-law shape of the X-ray afterglow and the reverse shock in the optical afterglow are the basic predictions (Paczynski & Rhoads 1993; Mészáros & Rees 1997; Sari & Piran 1999) of the classical hot fireball model of GRBs (see Appendices B.2 and B.3). Thus, the jets of these high-efficiency GRBs are likely to be thermal-dominated, and the radiation mechanism of the prompt emission is unlikely to be the ICMART model (for Poynting flux-dominated outflows; Zhang & Yan 2011). Also, considering the high efficiency ($\epsilon_\gamma \gtrsim 80\%$), the internal shock model ($\epsilon_\gamma \sim 10\%$; Rees & Meszaros 1994; Kobayashi et al. 1997) is unlikely. The prompt emission of these GRBs is likely to be produced by the photosphere emission, then.

In Figure 1(a), we show the correlation of L_{iso} and Γ for the selected GRBs. The Γ is obtained by the tight $L_{\text{iso}} - E_p - \Gamma$ correlation (Liang et al. 2015; the values are taken from Xue et al. 2019). In Section 4.3, we find that this estimation is likely to be quite accurate for these high-efficiency GRBs with $E_p \propto (L_{\text{iso}})^{1/4}$. Although four GRBs have detections of the peak time of the optical afterglow, we do not use them to estimate the Γ , due to the significant reverse shock signals. We

⁶ Interestingly, a similar X-ray afterglow characteristic has been found for GeV-/TeV-detected GRBs (Yamazaki et al. 2020).

Table 6The Observed Quantities, Inferred Efficiency (E_{ratio} and $(R_{\text{ph}}/R_s)^{-2/3}$), and Inferred Γ (from $T_{p,\text{op}}$ and the Prompt Emission) for Bursts with $T_{p,\text{op}}$ Detections

$\epsilon_\gamma \lesssim 50\%$ Subsample (24 Bursts)												
GRB	z	$dE_{p,z}/E_{p,z}$	$E_{p,z}$ (keV)	E_{iso} (10^{52} erg)	E_{ratio}	$(R_{\text{ph}}/R_s)^{-2/3}$	$T_{90,i}$ s	$T_{p,\text{op}}$ s	Γ_{op}	Γ_E	Γ_L	L_{iso} (10^{52} erg s $^{-1}$)
$dE_{p,z}/E_{p,z} \lesssim 0.2$												
060124	2.3	0.176	635.0 ± 112.0	43.000	0.640	0.362	72.2	631.0	220.0	272.0	451.0	14.200
090618	0.54	0.059	156.0 ± 9.2	25.300	0.267	0.710	67.7	91.2	319.0	128.0	340.0	2.050
090926A	2.11	0.028	908.0 ± 25.0	200.000	0.617	0.732	4.3	8.1	851.0	798.0	828.0	74.000
091020	1.71	0.037	507.0 ± 19.0	7.910	0.833	0.689	10.8	135.0	295.0	316.0	254.0	3.300
100728B	2.11	0.116	404.0 ± 47.0	7.240	0.716	1.230	3.9	33.9	462.0	360.0	216.0	1.860
100814A	1.44	0.093	344.0 ± 32.0	8.200	0.491	0.444	60.8	589.0	164.0	170.0	167.0	0.920
110213A	1.46	0.087	241.0 ± 21.0	6.400	0.332	0.299	11.9	324.0	199.0	207.0	284.0	2.090
120922A	3.1	0.167	156.0 ± 26.0	20.000	0.234	0.261	44.5	891.0	190.0	138.0	398.0	2.900
130420A	1.3	0.054	331.0 ± 18.0	7.190	0.653	0.599	45.7	356.0	190.0	176.0	109.0	0.350
130612A	2.01	0.172	186.0 ± 32.0	0.716	0.488	0.443	1.9	110.0	246.0	219.0	212.0	0.875
130831A	0.48	0.164	80.9 ± 13.3	0.757	0.158	0.163	11.9	724.0	93.3	92.0	185.0	0.296
131231A	0.64	0.042	288.0 ± 12.0	20.000	0.288	0.299	17.7	100.0	240.0	237.0	239.0	1.700
140629A	2.28	0.199	282.0 ± 56.0	6.000	0.418	0.444	7.9	151.0	293.0	246.0	298.0	2.700
$dE_{p,z}/E_{p,z} \gtrsim 0.2$												
050922C	2.2	0.266	417.0 ± 111.0	4.530	0.773	0.739	2.	132.0	401.0	408.0	619.0	19.000
060210	3.91	0.323	575.0 ± 186.0	41.500	0.568	0.408	51.9	676.0	248.0	280.0	316.0	5.960
060418	1.49	0.250	571.0 ± 143.0	12.800	0.832	0.738	41.4	151.0	244.0	255.0	187.0	1.890
060607A	3.08	0.348	575.0 ± 200.0	10.900	0.886	0.774	25.	178.0	271.0	284.0	191.0	2.000
070110	2.35	0.460	370.0 ± 170.0	5.500	0.618	0.200	26.4	1170.0	135.0	207.0	117.0	0.451
081007	0.53	0.246	61.0 ± 15.0	0.170	0.178	1.090	6.5	123.0	152.0	77.1	85.8	0.043
091029	2.75	0.287	230.0 ± 66.0	7.400	0.297	0.317	10.5	407.0	218.0	213.0	234.0	1.320
100906A	1.73	0.348	158.0 ± 55.0	33.400	0.220	1.220	33.1	100.0	396.0	160.0	366.0	2.450
130215A	0.6	0.408	248.0 ± 101.0	2.500	0.471	0.447	90.	741.0	111.0	113.0	63.9	0.084
141109A	2.93	0.404	750.0 ± 303.0	31.000	0.891	0.150	23.5	955.0	193.0	376.0	240.0	4.200
141221A	1.45	0.228	372.0 ± 85.0	1.900	0.888	0.723	9.7	110.0	216.0	233.0	142.0	0.700
$\epsilon_\gamma \gtrsim 50\%$ Subsample (23 Bursts)												
GRB	z	$dE_{p,z}/E_{p,z}$	$E_{p,z}$ (keV)	E_{iso} (10^{52} erg)	E_{ratio}	η/Γ	$T_{90,i}$ s	$T_{p,\text{op}}$ s	Γ_{op}	Γ_E	Γ_L	L_{iso} (10^{52} erg s $^{-1}$)
990123	1.6	0.043	2030.0 ± 88.0	239.000	1.700	1.150	23.9	47.9	656.0	677.0	417.0	35.300
060605	3.78	0.512	490.0 ± 251.0	2.830	1.120	3.610	16.6	479.0	200.0	209.0	146.0	0.951
061121	1.31	0.093	1290.0 ± 120.0	26.100	1.940	10.300	7.7	162.0	301.0	543.0	332.0	14.100
071112C	0.82	0.546	597.0 ± 326.0	1.600	1.760	1.950	2.9	178.0	187.0	333.0	89.8	0.400
080319B	0.94	0.013	1260.0 ± 17.0	150.000	1.050	0.134	23.	17.4	810.0	508.0	281.0	9.590
080804	2.2	0.056	809.0 ± 45.0	11.500	1.370	0.443	10.6	63.1	437.0	358.0	189.0	2.690
080810	3.35	0.121	1490.0 ± 180.0	39.100	2.060	1.330	24.4	117.0	453.0	460.0	257.0	9.270
080916C	4.35	0.089	2760.0 ± 246.0	560.000	1.930	0.035	11.5	6.2	2060.0	1050.0	600.0	104.000
081203A	2.1	1.200	1540.0 ± 1854.0	35.000	2.240	3.020	96.4	309.0	274.0	327.0	146.0	2.810
090323	3.57	0.108	1900.0 ± 206.0	390.000	1.320	3.580	29.1	200.0	504.0	663.0	446.0	38.500
090812	2.45	0.449	2020.0 ± 908.0	40.300	3.060	0.496	9.5	47.9	583.0	681.0	229.0	9.550
090902B	1.82	0.015	2020.0 ± 31.0	440.000	1.380	0.094	6.9	8.5	1390.0	994.0	529.0	58.900
100414A	1.39	0.020	1490.0 ± 29.0	76.900	1.640	0.281	9.2	34.7	622.0	638.0	226.0	7.000
110731	2.83	0.057	1210.0 ± 69.0	40.000	1.550	0.041	1.75	5.0	1410.0	805.0	459.0	27.000
120711A	1.4	0.039	2340.0 ± 91.0	150.000	2.400	7.910	17.2	240.0	328.0	744.0	266.0	15.200
120909A	3.93	0.067	1820.0 ± 123.0	72.900	2.190	2.400	23.3	288.0	366.0	557.0	210.0	7.190
130427A	0.34	0.009	1380.0 ± 13.0	80.900	1.450	1.260	46.1	21.9	599.0	413.0	435.0	27.000
130610A	2.09	0.419	912.0 ± 382.0	6.810	1.920	3.150	7.	204.0	260.0	395.0	171.0	2.400
160629A	3.33	0.074	1280.0 ± 95.0	47.000	1.580	0.688	14.9	81.3	531.0	494.0	272.0	9.100
061007	1.26	0.030	902.0 ± 27.0	88.100	0.804 ^a	2.220	25.5	74.1	466.0	392.0	427.0	17.400
110205A	2.22	0.335	714.0 ± 239.0	56.000	0.685 ^a	8.540	79.8	813.0	205.0	248.0	193.0	2.500
121128A	2.2	0.050	198.0 ± 10.0	14.000	0.197 ^a	1.220	3.1	74.1	422.0	247.0	516.0	6.400
140423A	3.26	0.122	533.0 ± 65.0	56.000	0.464 ^a	1.550	22.3	200.0	385.0	294.0	319.0	5.660

Note.^a $\epsilon_\gamma \gtrsim 50\%$ judged from E_{iso}/E_k .

Table 7
The Observed Quantities and Properties of $E_{\text{iso}}/L_{\text{X},11}$ for Bursts with Maximum Γ

GRB	z	$T_{90,i}$ (s)	$dE_{p,z}/E_{p,z}$	E_{ratio}	$E_{p,z}$ (keV)	E_{iso} (10^{52} erg)	$L_{\text{X},11}$ (10^{45} erg s $^{-1}$)	$E_{\text{iso}}/L_{\text{X},11}$	$T_{p,\text{op}}$ (s)	Γ_{op}	$\Gamma_{\text{op}}/\Gamma_{\text{max}}$	L_{iso} (10^{52} erg s $^{-1}$)
100414A	1.39	9.2	0.020	1.640	1490.0 ± 29.0	76.900			34.7	622.0	1.040	7.000
090812	2.45	9.5	0.449	3.060	2020.0 ± 908.0	27.18	6.525	4.166	47.9	583.0	0.902	9.550
160629A	3.33	14.9	0.074	1.580	1280.0 ± 95.0	47.000			81.3	531.0	0.832	9.100
100728B	2.11	3.9	0.116	1.000	404.0 ± 47.0	2.66	0.841	3.163	33.9	462.0	1.076	1.860
050502A	3.79			1.025	498.9	3.981			57.5	461.4	0.989	2.600
080804	2.2	10.6	0.056	1.370	809.0 ± 45.0	11.500	3.181	3.615	63.1	437.0	0.928	2.690
081008	1.97			0.438	267.3	4.19	0.915	4.579	162.2	261.5	0.961	0.300
080310	2.42			0.087	75.0	3.25	1.78		182.0	255.8	0.881	0.390
									(2000)	(104)	(0.352)	
081007	0.53	6.5	0.246	0.178	61.0 ± 15.0	0.170			123.0	152.0	0.908	0.043

find that Γ is tightly correlated with L_{iso} , $\Gamma \propto (L_{\text{iso}})^{0.29}$. This is consistent with the prediction of the neutrino annihilation from the hyperaccretion disk (Lü et al. 2012), $\Gamma \propto (L_{\text{iso}})^{7/27} = (L_{\text{iso}})^{0.26}$. This therefore also supports the jets of these GRBs being thermal-dominated.

In Figure 1(b), we show the correlation of E_{iso}/E_k and η/Γ (see Section 2.2.1) for the selected GRBs. According to Equation (10), along with $r_0 \sim 3.21 \times 10^8$ cm, as derived above, and L_{iso} , we can use Γ to obtain the η for each burst. We find obvious linear correlations for E_{iso}/E_k and η/Γ , and they are almost the same (aside from three bursts: note that we obtain four bursts that are almost the same when we take $r_0 = 3.21 \times 10^8$ cm, and then use the offset from the best-fit $E_p \propto (L_{\text{iso}})^{1/4}$ relation to slightly modify r_0 for the other bursts, to obtain three other bursts that are almost the same) when we take $E_{k,52} = 5 * L_{\text{X},45}$ ($E_{k,52} = E_k/10^{52}$, $L_{\text{X},45} = L_{\text{X}}/10^{45}$: this is quite close to the derivation of $E_{k,52} = 3.7 * L_{\text{X},45}$ described in Section 4.1, and the slight difference is likely to result from the slight error of Γ estimated by the $L_{\text{iso}} - E_p - \Gamma$ correlation). Again, this result strongly supports photosphere emission origin in the unsaturated acceleration regime for these high-efficiency GRBs.

3.3. Discussion of the Probability Photosphere Model and the Dissipative Photosphere Model

According to the above statements, the prompt emission of the selected high-efficiency GRBs is likely to be produced by photosphere emission in the unsaturated acceleration regime. In Table 1, however, it is noteworthy that the low-energy spectral index α is quite typical (around -1), rather than very hard. This strongly supports the photosphere emission model having the capacity to produce the observed typical soft low-energy spectrum. Theoretically, the probability photosphere model (with geometric broadening) and the dissipative photosphere model (with subphotospheric energy dissipation) can both achieve this. But for the dissipative photosphere model, the $E_p \propto (L_{\text{iso}})^{1/4}$ relation should be violated, since inverse Compton scattering below the photosphere radius will change the photon energy (namely $T_{\text{ob}} \neq T_0$; see Appendix A). Also, the high-energy spectrum for this model should be a power law, rather than the exponential cutoff. So the characteristics of the selected high-efficiency GRBs favor the probability photosphere model.

4. Evidence from Long GRBs with $\epsilon_\gamma \gtrsim 50\%$ and $\epsilon_\gamma \lesssim 50\%$

4.1. $\epsilon_\gamma = 50\%$ and Maximum Γ

For the $\epsilon_\gamma = 50\%$ ($R_{\text{ph}} = R_s$) case, with a fixed L_{iso} , the observed Lorentz factor Γ in the afterglow phase should be the maximum, for the following reason. To obtain $\epsilon_\gamma < 50\%$ ($R_{\text{ph}} > R_s$, $R_{\text{ph}} \propto L_{\text{iso}}/\Gamma^3$, and $R_s = \Gamma \cdot r_0$), Γ ($\Gamma = \eta$) should be smaller. Conversely, for $\epsilon_\gamma > 50\%$, η should be larger. And in this case, from Equation (10), we have $\Gamma \propto (L_{\text{iso}}/\eta)^{1/3}$, thus Γ should also be smaller. Note that this maximum Γ exists for the hot fireball, while the corresponding $\epsilon_\gamma = 50\%$ is the prediction of the photosphere emission origin for the prompt emission. The maximum Γ is given as (see also Equation (16) in Ghirlanda et al. 2018)

$$\Gamma_{\text{max}} = \left[\frac{L_{\text{iso}} \sigma_T}{8\pi m_p c^3 r_0} \right]^{1/4}. \quad (15)$$

In Figure 11(a), we show the distribution of L_{iso} and Γ for the complete sample (62 bursts), with the detection of the peak time of the early optical afterglow (Ghirlanda et al. 2018; obtaining Γ). Obviously, with the exception of GRB 080319B (with a strong reverse shock signal) and four bursts—the peak time is obtained from the Fermi/Large Area Telescope (LAT) light curve, and the decay slope of ~ 1.5 implies that it is likely to be produced by the radiative fireball, so the Γ should be smaller by a factor of ~ 1.6 (see Ghisellini et al. 2010 and Appendix B.4—the distribution of the maximum Γ well follows the predicted $L_{\text{iso}}^{1/4}$ correlation, and only has the difference of a constant $\sim 10^{0.1}$ (1.27) from the prediction for Equation (15) (the dashed line, $r_0 \sim 3.21 \times 10^8$ cm, is based on Figure 3). Note that although the equation for calculating Γ is confirmed to act as $\Gamma \propto (E_k)^{1/8} \cdot [T_p/(1+z)]^{-3/8}$, its constant is highly uncertain (see Table 2 in Ghirlanda et al. 2018). The constant that is given in other works (with different methods) can be 1.7 (or 0.5) times that used in Ghirlanda et al. (2018). So the above difference (1.27) obtained by our work is reasonable, and may be more accurate (since it does not strongly depend on the model assumption; if r_0 is accurate, then it is likely to be accurate).

We then select the sample (nine bursts) with the maximum Γ (see Table 7) to check their efficiency properties. In Figure 11(b), we show the distribution of E_{iso} and $L_{\text{X},45}$ for this sample (four bursts with $L_{\text{X},45}$ detection). Note that we exclude GRB 081007, due to the too small E_{iso} , and GRB 080310, due to the plateau in the early optical afterglow. As expected from the photosphere emission model, all these bursts

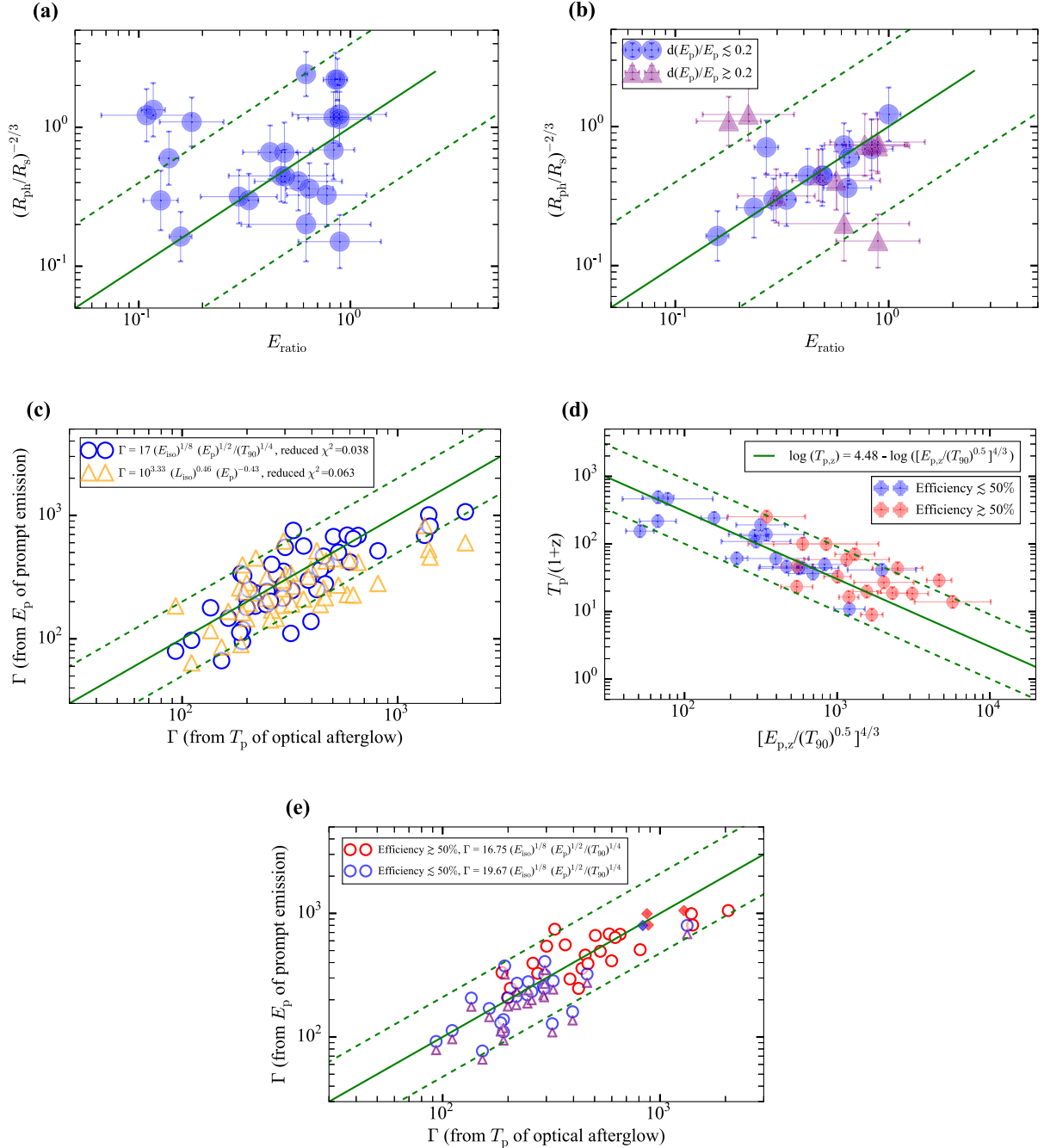


Figure 7. The tight correlation of $\Gamma \propto E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4}$ to estimate Γ , which we derived using the same efficiency of E_{ratio} and $(R_{\text{ph}}/R_s)^{-2/3}$ as in the $\epsilon_\gamma \lesssim 50\%$ case (see Table 6). (a) The distribution of E_{ratio} and $(R_{\text{ph}}/R_s)^{-2/3}$ for the selected $\epsilon_\gamma \lesssim 50\%$ sample (based on $E_{\text{ratio}} \leq 0.9$; 24 bursts; reduced $\chi^2 = 0.239$). (b) The distributions of E_{ratio} and $(R_{\text{ph}}/R_s)^{-2/3}$ for the two subsamples for smaller E_p errors ($dE_p/E_p \leq 0.2$; blue circles; reduced $\chi^2 = 0.025$) and larger E_p errors ($dE_p/E_p \geq 0.2$; purple triangles; reduced $\chi^2 = 0.228$), respectively. (c) A comparison of the Γ obtained from the optical afterglow (for 47 bursts, from Ghirlanda et al. 2018) and the Γ obtained from the prompt emission (orange triangles for $\Gamma = 10^{3.33} L_{\text{iso}}^{0.46} E_p^{-0.43}$ and blue circles for $\Gamma = 17 \cdot E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4}$). (d) The distribution of $[E_p/(T_{90})^{1/2}]^{4/3}$ and $T_p/(1+z)$ for 35 bursts (reduced $\chi^2 = 0.100$). (e) The slightly different constants for the $\epsilon_\gamma \lesssim 50\%$ case (Const = 19.67, reduced $\chi^2 = 0.034$; Const = 16.75, purple triangles, reduced $\chi^2 = 0.048$) and the $\epsilon_\gamma \gtrsim 50\%$ case (Const = 16.75).

have almost the same efficiency (with $E_{\text{iso}} \propto L_X \propto E_k$). Thus, we think that the efficiency ϵ_γ for these bursts is likely to be 50% ($E_{\text{iso},52} = E_{k,52} \simeq 3.7 \cdot L_{X,45}$). Note that, based on this, the average derived efficiency ($\epsilon_\gamma \sim 33\%$ to 40% ; see Figure 10) for the whole sample (117 bursts) is almost consistent with that given in other works (Lloyd-Ronning & Zhang 2004; Fan & Piran 2006; Zhang et al. 2007; D’Avanzo et al. 2012; Wygoda et al. 2016).

4.2. $E_p - E_{\text{iso}}$ Distributions and Consistent Efficiency

Based on the derivation of $E_{k,52} = 3.7 \cdot L_{X,45}$ described above, and separated by $E_{\text{iso},52} = E_{k,52} = 3.7 \cdot L_{X,45}$ ($\epsilon_\gamma = 50\%$) for the distribution of $E_{\text{iso},52}$ and $L_{X,45}$ in Figure 11, we obtain two distinguished long-GRB samples ($\epsilon_\gamma \lesssim 50\%$, see Table 2; and $\epsilon_\gamma \gtrsim 50\%$, see Table 3). Note that we exclude the above high-efficiency sample ($\epsilon_\gamma \gtrsim 80\%$) and the sample with $\epsilon_\gamma = 50\%$.

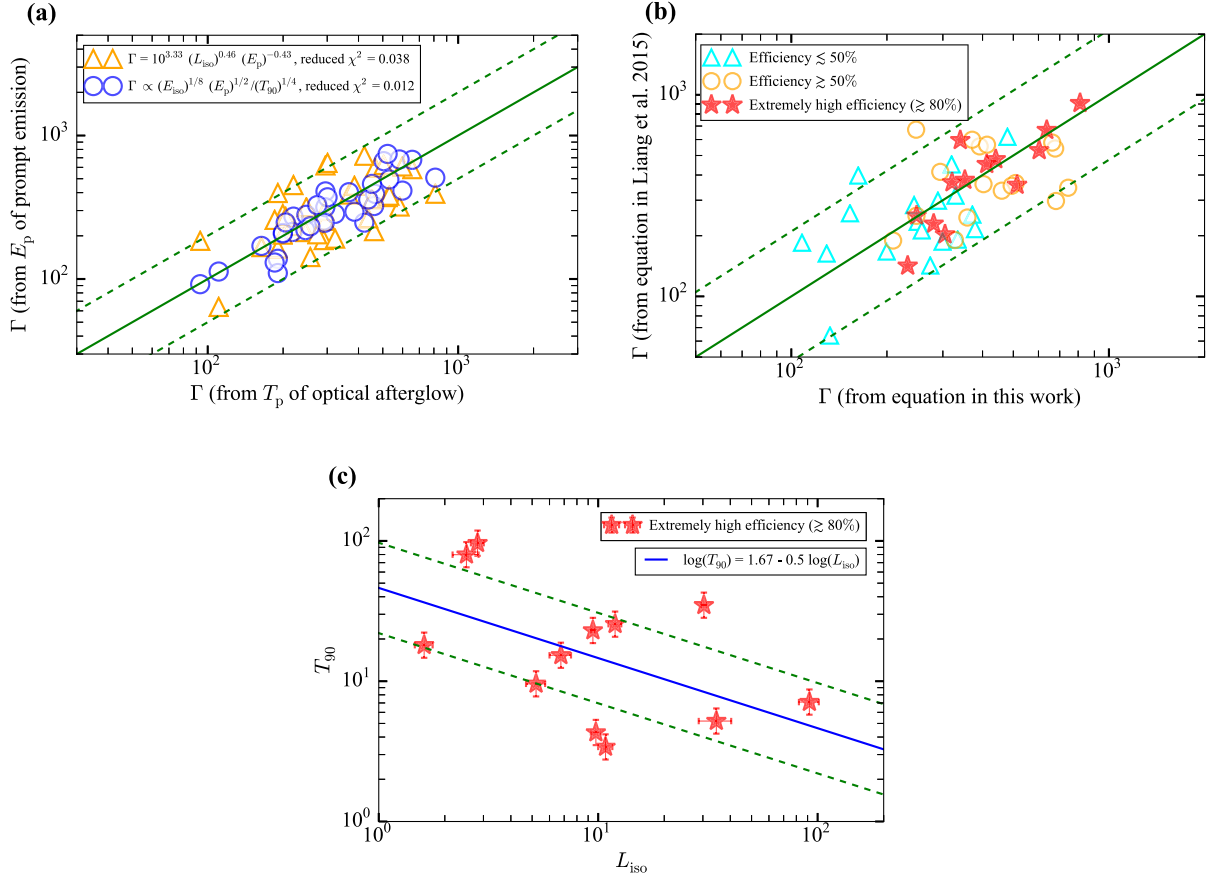


Figure 8. Comparison of Γ as obtained from $\Gamma \propto E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4}$ and from $\Gamma = 10^{3.33} L_{\text{iso}}^{0.46} E_p^{-0.43}$ (see Tables 6 and 1). (a) Comparison of the Γ obtained from the optical afterglow and the Γ obtained from the prompt emission (orange triangles for $\Gamma = 10^{3.33} L_{\text{iso}}^{0.46} E_p^{-0.43}$ and blue circles for $\Gamma \propto E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4}$ for the better sample (35 bursts) in Figure 7(d). The different constants are used for the $\epsilon_\gamma \gtrsim 50\%$ and $\epsilon_\gamma \lesssim 50\%$ subsamples. (b) Comparison of Γ as obtained from $\Gamma = 10^{3.33} L_{\text{iso}}^{0.46} E_p^{-0.43}$ and from $\Gamma \propto E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4}$ for the $\epsilon_\gamma \lesssim 50\%$ (cyan triangles; reduced $\chi^2 = 0.043$), the $\epsilon_\gamma \gtrsim 50\%$ (orange circles; reduced $\chi^2 = 0.046$), and the high-efficiency ($\epsilon_\gamma \gtrsim 80\%$; red stars; reduced $\chi^2 = 0.017$) subsamples. Obviously, these two estimations are consistent with each other, since they can be transferred with $E_p \propto (E_{\text{iso}})^{1/4}$. (c) The $T_{90} \propto (L_{\text{iso}})^{-0.5}$ correlation (reduced $\chi^2 = 0.174$) found for the high-efficiency subsample ($\epsilon_\gamma \gtrsim 80\%$).

In Figure 4(a), we show that the best-fit result for the $\epsilon_\gamma \gtrsim 50\%$ sample is $\log(E_p) = 2.47 + 0.25 \log(E_{\text{iso}})$, which is consistent with the prediction of the photosphere emission model. Also, this result is almost the same as that for the above high-efficiency sample. The offset from the best-fit result is likely to be caused by a distribution of r_0 (as for the constrained results in Pe'er et al. 2015). For the $\epsilon_\gamma \lesssim 50\%$ sample, E_{iso} and E_p should both decrease by the same factor of $(R_{\text{ph}}/R_s)^{-2/3}$ (see Section 2.2.3), compared with the distribution of $\log(E_p) = 2.54 + 0.25 \log(E_{\text{iso}})$ for the above high-efficiency sample. In Figure 4(a), we show that the upmost distribution for the $\epsilon_\gamma \lesssim 50\%$ sample is around $\log(E_p) = 2.54 + 0.25 \log(E_{\text{iso}})$, and that the best-fit result (much smaller) is $\log(E_p) = 2.31 + 0.26 \log(E_{\text{iso}})$. The decreases of E_{iso} and E_p are more obvious when we divide the $\epsilon_\gamma \lesssim 50\%$ sample into two subsamples ($\epsilon_\gamma \lesssim 17\%$ and $\epsilon_\gamma \gtrsim 17\%$).

In Figure 2(a), we show the distributions of E_{ratio} ($E_{\text{ratio}} = [(E_p/2.7k)^4 (4\pi r_l^2 ac)/E_{\text{iso}}]^{1/3}$) and E_{iso}/E_k (implicitly, $E_{k,52} = 3.7 L_{X,45} / ((1+z)/2)$ is adopted) for the $\epsilon_\gamma \lesssim 50\%$ sample. They are centered around the equal-value line and have a linear correlation. This is consistent with the prediction from the photosphere emission model (see Section 2.2.4). The dispersion is likely to be caused by the estimation error for E_k (we have excluded the bursts with large E_p errors of $dE_p/E_p \geq 0.2$), since many X-ray afterglow light

curves are quite complex (with plateaus, steep decays, or significant flares). The method of using $L_{X,11h}$ to estimate E_k should only be completely correct for X-ray afterglows with power-law shapes and slopes of -1 . To check the origin of the dispersion, we select the bursts with almost the same E_{ratio} and E_{iso}/E_k (see Table 5). As expected, we find that all these bursts (seven bursts) have a power-law X-ray afterglow light curve with a slope of ~ -1 (shown in Figures 2(b) and (c)).

Also, according to Section 2.2.4, for the $\epsilon_\gamma \lesssim 50\%$ sample, we should have $E_{\text{ratio}} = E_{\text{iso}}/E_k = (R_{\text{ph}}/R_s)^{-2/3}$. To check this, we select the bursts with detections of the peak time of the optical afterglow (using them to estimate Γ and thus $(R_{\text{ph}}/R_s)^{-2/3}$). Note that since E_{ratio} and E_{iso}/E_k are the average results for the whole duration, and $(R_{\text{ph}}/R_s)^{-2/3} \propto L^{-2/3}$, we use $L = E_{\text{iso}}/T_{90}$ (rather than L_{iso}). In Figure 2(d), we show the distributions of E_{ratio} , E_{iso}/E_k , and $(R_{\text{ph}}/R_s)^{-2/3}$ for the selected sample (six bursts). Similar to the above, they are centered around the equal-value line and have a linear correlation. Also, one burst has almost the same values for these three quantities, and the five other bursts have almost the same values for two quantities. So the predicted $E_{\text{ratio}} = E_{\text{iso}}/E_k = (R_{\text{ph}}/R_s)^{-2/3}$ from the photosphere emission model can be well reproduced.

For the $\epsilon_\gamma \gtrsim 50\%$ sample, similar to Figure 1(b) (for high-efficiency GRBs), we should have $E_{\text{iso}}/E_k = \eta/\Gamma$. To check this, we also select the bursts with detections of the peak time

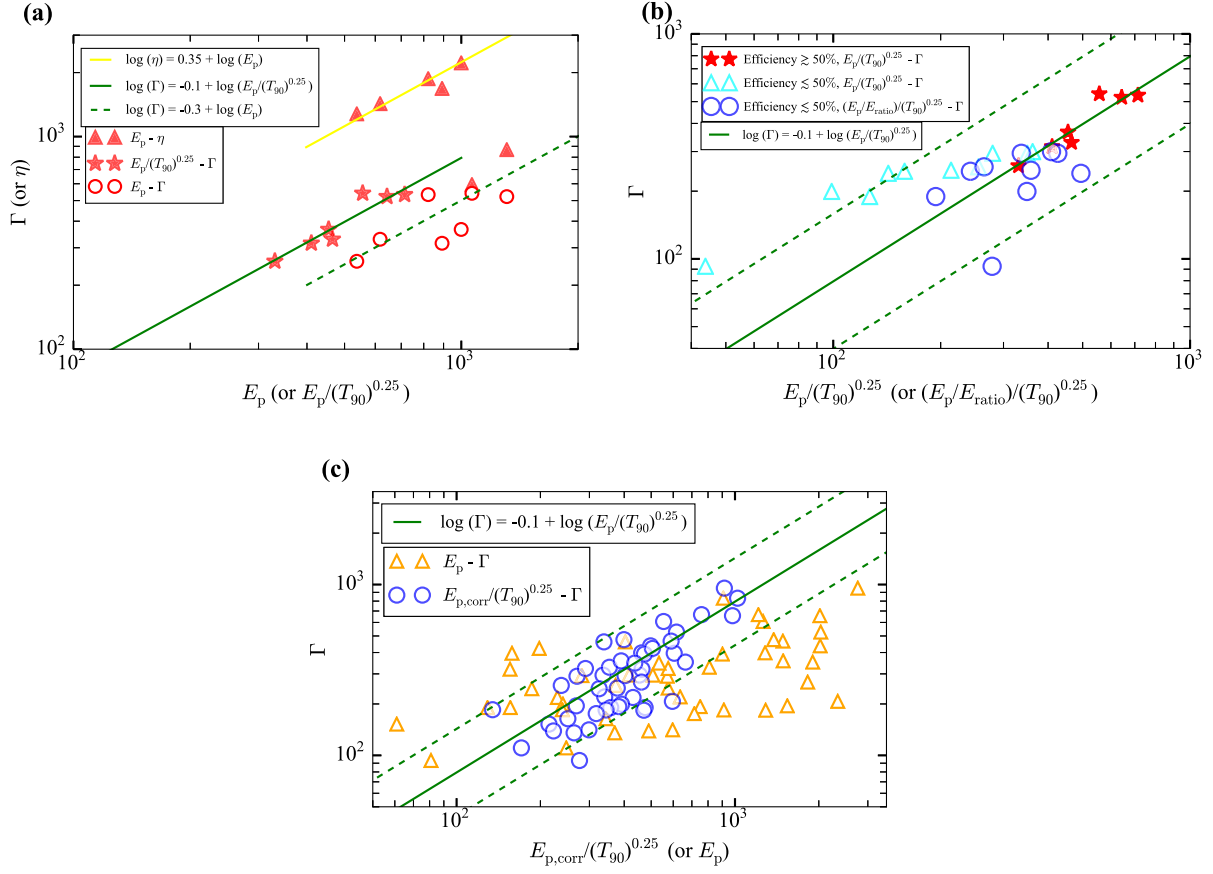


Figure 9. The tight correlation of $\Gamma \propto E_p/(T_{90})^{1/4}$, predicted by $\Gamma \propto E_{\text{iso}}^{1/8} E_p^{1/2}/(T_{90})^{1/4}$ and $E_p \propto (E_{\text{iso}})^{1/4}$. (a) The distributions of $E_p - \Gamma$ (red circles; reduced $\chi^2 = 0.013$), $E_p - \eta$ (red triangles; reduced $\chi^2 = 0.002$), and $E_p/(T_{90})^{1/4} - \Gamma$ (red stars; reduced $\chi^2 = 0.002$) for the selected $\epsilon_\gamma \gtrsim 50\%$ sample with T_p detections (seven bursts in Figure 2(e); see Table 3). It is obvious that the tight correlations of $\Gamma = 10^{-0.1} \cdot E_p/(T_{90})^{1/4}$ and $E_p \propto \eta$ (for the five bursts with higher efficiency, thus $\eta \propto (E_{\text{iso}})^{1/4}$) are found. (b) The distributions of $(E_p/E_{\text{ratio}})/(T_{90})^{1/4} - \Gamma$ (blue circles; reduced $\chi^2 = 0.031$) and $E_p/(T_{90})^{1/4} - \Gamma$ (cyan triangles) for the selected $\epsilon_\gamma \lesssim 50\%$ sample with T_p detection (six bursts in Figure 2(d); see Table 2). The tight correlation of $(E_p/E_{\text{ratio}})/(T_{90})^{1/4} \propto \Gamma$ is found, which is in line with the $E_p/(T_{90})^{1/4} \propto \Gamma$ correlation for the $\epsilon_\gamma \gtrsim 50\%$ case. (c) The distribution of $\Gamma - E_p/(T_{90})^{1/4}$ (blue circles; reduced $\chi^2 = 0.025$) for the large sample with Γ (47 bursts; see Table 6) in Ghirlanda et al. (2018). For the $\epsilon_\gamma \gtrsim 50\%$ case, the E_p is rederived from the $\log(E_p) = 2.54 + 0.25 \log(E_{\text{iso}})$ correlation (using E_{iso}), and for the $\epsilon_\gamma \lesssim 50\%$ case, the E_p is rederived from E_p/E_{ratio} . Obviously, we find that the distribution of Γ and $E_p/(T_{90})^{1/4}$ is centered around $\Gamma = 10^{-0.1} \cdot E_p/(T_{90})^{1/4}$ and has a linear correlation.

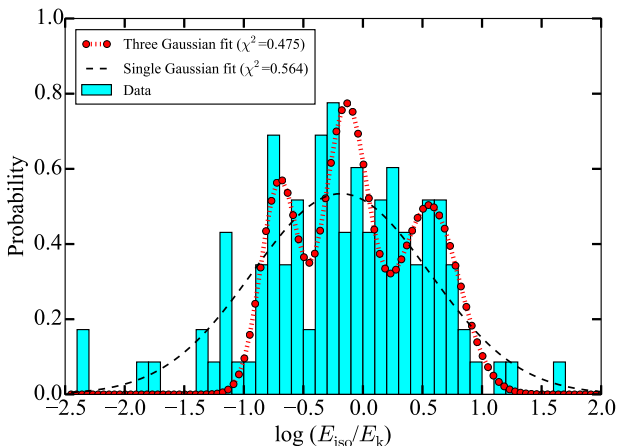


Figure 10. The distribution of E_{iso}/E_k for our whole sample (117 bursts, after GRB 110213A; see Tables 1–3). The mean value is around $\sim 10^{-0.2} - 10^{-0.3}$, thus indicating the average efficiency of $\epsilon_\gamma \sim 33\% - 40\%$. The distribution also seems to consist of three Gaussian distributions (smaller χ^2 for $E_{\text{iso}}/E_k \gtrsim 10^{-1.0}$).

of the optical afterglow (using them to estimate Γ and thus η). Note that since $\Gamma \propto L^{1/3}$ and the derived Γ is likely to correspond to L_{iso} (the maximum L), we use $L = L_{\text{iso}}$ when calculating η . In Figure 2(e), we show the distribution of E_{iso}/E_k and η/Γ for the selected sample (seven bursts). As expected, they are centered around the equal-value line and have a linear correlation.

The analysis results obtained above are for the sample with $L_{X,11h}$ as derived in this work (for GRBs after GRB 110213A). For another sample, with $L_{X,11h}$ as presented in D’Avanzo et al. (2012; for GRBs before GRB 110213A), we perform a similar analysis and obtain similar results. For the $\epsilon_\gamma \gtrsim 50\%$ subsample, the E_p and E_{iso} distribution is also centered around $\log(E_p) = 2.47 + 0.25 \log(E_{\text{iso}})$ (see Figure 4(b)). For the $\epsilon_\gamma \lesssim 50\%$ subsample, the decreases of E_{iso} and E_p are also obvious. Also, the distributions of E_{ratio} and E_{iso}/E_k are centered around the equal-value line and have linear correlations (see Figure 6(a)). For the selected bursts, with almost same E_{ratio} and E_{iso}/E_k (see Table 5), all (seven bursts) show a power-law X-ray afterglow light curve with a slope of ~ -1 (see Figures 6(b) and (c)).

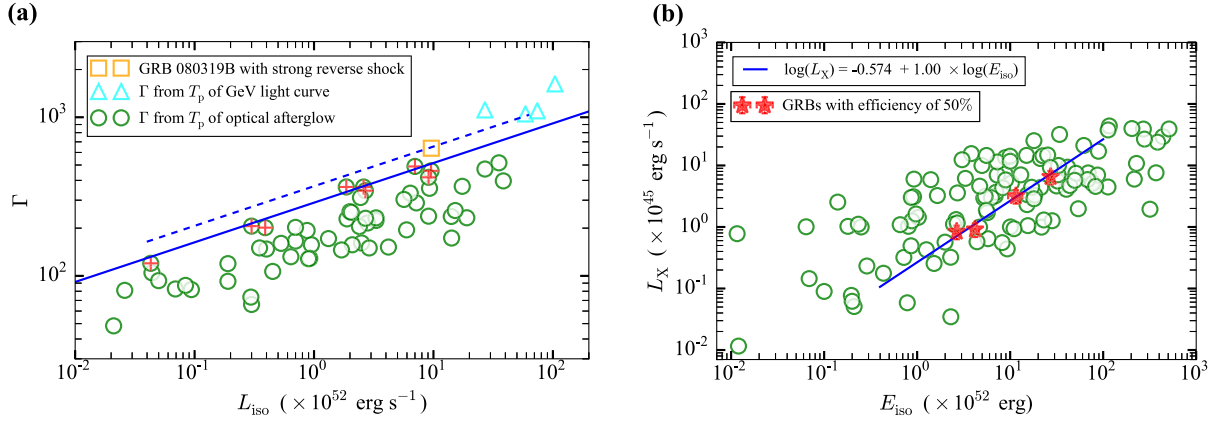


Figure 11. The $L_{\text{iso}} - \Gamma$ and $E_{\text{iso}} - L_{X,45}$ distributions for the $\epsilon_\gamma = 50\%$ sample (see Table 7). (a) The distribution of L_{iso} and Γ for the complete sample (62 bursts) with the detection of T_p . We select the sample (nine bursts, marked by the red pluses) with the maximum Γ (for fixed L_{iso}) to check whether their efficiency is 50%, as predicted by the photosphere emission model. (b) The distribution of E_{iso} and $L_{X,45}$ for the selected sample with the maximum Γ (four bursts with $L_{X,45}$ detection, marked by the red stars). It is found that all these bursts have almost the same efficiency (with $E_{\text{iso}} \propto L_X \propto E_k$, reduced $\chi^2 = 0.008$). Thus, we think that the efficiency ϵ_γ for these bursts is likely to be 50% ($E_{\text{iso},52} = E_{k,52} \simeq 3.7 * L_{X,45}$).

4.3. The Excellent Derived $\Gamma \propto E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4}$ Correlation

The small burst number in Figure 2(d) is a result of obtaining both the X-ray afterglow light curve and the detection of the peak time of the early optical afterglow. To further check $E_{\text{ratio}} = E_{\text{iso}}/E_k = (R_{\text{ph}}/R_s)^{-2/3}$ for the $\epsilon_\gamma \lesssim 50\%$ case, we then analyze the complete sample with detections of the peak time of the early optical afterglow (obtaining Γ and thus $(R_{\text{ph}}/R_s)^{-2/3}$; Ghirlanda et al. 2018; see Table 6). Note that the constant used is 1.27 times that given in Ghirlanda et al. (2018; see Section 4.1). Although E_{iso}/E_k is missing from most bursts in the sample, considering the different distributions of E_p and E_{iso} for $\epsilon_\gamma \lesssim 50\%$ and $\epsilon_\gamma \gtrsim 50\%$, we can use the judgment of $(E_{\text{ratio}} \lesssim 0.9)$ to roughly select the $\epsilon_\gamma \lesssim 50\%$ subsample. Note that we do not use the bursts without the E_p value in Minaev & Pozanenko (2020) and Xue et al. (2019), due to the large E_p errors, and we move four bursts to the $\epsilon_\gamma \gtrsim 50\%$ subsample, based on their detections of E_{iso}/E_k . In Figure 7(a), we show the distribution of E_{ratio} and $(R_{\text{ph}}/R_s)^{-2/3}$ for the selected $\epsilon_\gamma \lesssim 50\%$ subsample (24 bursts). This distribution is roughly centered around the equal-value line, and has a linear correlation. After modifying Γ or E_p , based on Figure 2(d) (using $E_{\text{ratio}} = E_{\text{iso}}/E_k = (R_{\text{ph}}/R_s)^{-2/3}$) for the six bursts there, in Figure 7(b), we show the distributions of E_{ratio} and $(R_{\text{ph}}/R_s)^{-2/3}$ for the two subsamples with smaller E_p errors ($dE_p/E_p \leq 0.2$) and larger E_p errors ($dE_p/E_p \geq 0.2$), respectively. Obviously, for the subsample with smaller E_p errors, the values of E_{ratio} and $(R_{\text{ph}}/R_s)^{-2/3}$ are almost the same. Note that we decrease the Γ of GRB 090926A by a factor of 1.6, since its peak time is obtained from the LAT light curve and the decay slope of ~ 1.5 implies that it is likely to be produced by the radiative fireball (Ghisellini et al. 2010). And, for the large offset in Figure 7(b) (the upper offset), we check its (GRB 090618) optical afterglow light curve, and find that the reverse shock signal is significant, thus overestimating Γ and $(R_{\text{ph}}/R_s)^{-2/3}$.

Based on $E_{\text{ratio}} = (R_{\text{ph}}/R_s)^{-2/3}$, we derive $\Gamma \propto E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4}$ (see Equation (6) and Equation (7) in Section 2.2.5). In Figure 7(c), we show a comparison of the Γ obtained from the optical afterglow (for 47 bursts in Ghirlanda et al. 2018) and the Γ obtained from the prompt emission (orange triangles for $\Gamma = 10^{3.33} L_{\text{iso}}^{0.46} E_p^{-0.43}$ and blue circles for $\Gamma = 17 \cdot E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4}$). Obviously, using these two correlations, we can give an approximate estimations for both Γ . Furthermore,

Equation (7), as derived in our work from the photosphere emission model, can give a better estimation for Γ (with a smaller reduced χ^2).

According to Equation (7), two correlations of $T_p/(1+z) \propto [E_p/(T_{90})^{1/2}]^{-4/3}$ (see Section 2.2.6) and $\Gamma \propto E_p/(T_{90})^{1/4}$ (see Section 2.2.7) are predicted, which will be tested in the following section.

4.3.1. $T_p/(1+z) \propto [E_p/(T_{90})^{1/2}]^{-4/3}$ Test

In Figure 7(d), we show the distribution of $E_p/(T_{90})^{1/2}$ and $T_p/(1+z)$ for 35 bursts (we select seven bursts with $dE_p/E_p \geq 0.2$ and five bursts with LAT light curves, and we modify the E_p or T_p for five bursts, based on Figure 2(d)). Just as predicted, this distribution shows a linear correlation with a slope of $\sim -4/3$, and it is consistent with $T_p/(1+z) \propto (E_p)^{-1.25}$ presented in Table 1 of Ghirlanda et al. (2018). So Equation (7) is likely to be correct.

Besides, from Figure 7(d), we can see that the T_p for the $\epsilon_\gamma \gtrsim 50\%$ subsample is a bit larger than that for $\epsilon_\gamma \lesssim 50\%$, although both satisfy the $T_p/(1+z) \propto [E_p/(T_{90})^{1/2}]^{-4/3}$ correlation. This is consistent with the slightly different predicted constants from Equation (7) for the $\epsilon_\gamma \lesssim 50\%$ case (19.67; see Equation (8)) and the $\epsilon_\gamma \gtrsim 50\%$ case (16.75; see Equation (11)).

The smaller constant (16.75) for the $\epsilon_\gamma \gtrsim 50\%$ case is consistent with the larger T_p in Figure 7(d). In Figure 7(e), we show the derived Γ using Equation (8) and Equation (11), and find that the distribution for the $\epsilon_\gamma \lesssim 50\%$ case is better (more symmetric) than that in Figure 7(c).

4.3.2. $\Gamma = 10^{-0.1} \cdot E_p/(T_{90})^{1/4}$ Correlation

In Figure 9(a), we find the tight correlation of

$$\Gamma = 10^{-0.1} \cdot E_p/(T_{90})^{1/4} \quad (16)$$

for the $\epsilon_\gamma \gtrsim 50\%$ case (with $L_{X,11h}$ detection). Note that the E_p here is rederived from the $\log(E_p) = 2.54 + 0.25 \log(E_{\text{iso}})$ correlation (using E_{iso}), since the observed E_p with an offset from the above line is likely to arise from the different r_0 (actually $E_p \propto (E_{\text{iso}}/r_0^2)^{1/4}$) or the error of E_p . Besides, we modify Γ (using E_{iso}/E_k , mainly for three bursts) based on

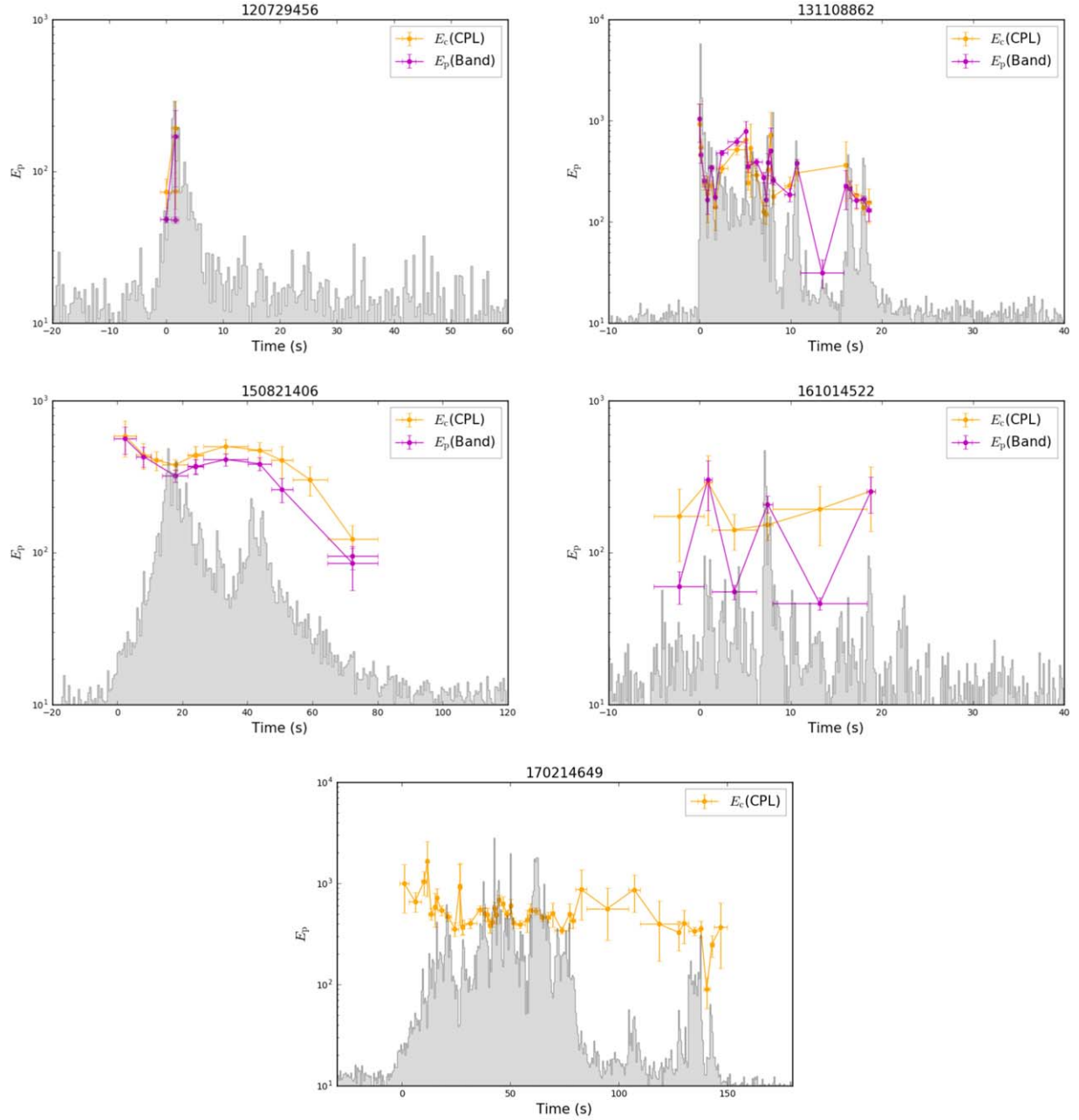


Figure 12. The E_p evolutions of the time-resolved spectra for five extremely high-efficiency GRBs ($\epsilon_\gamma \gtrsim 80\%$) detected by Fermi/GBM. The E_p evolutions almost follow the evolution of the flux F , consistent with the $E_p \propto F^{1/4}$ predicted by the photosphere (thermal) emission in the unsaturated acceleration regime.

Figure 2(e). In Figure 9(a), we also show the distributions for $E_p - \Gamma$ and $E_p - \eta$, where η is obtained by $\eta/\Gamma = E_{\text{iso}}/E_k$. It is obvious that there is a tight correlation of $E_p \propto \eta$ for the subsample (five bursts) with higher efficiency. This means that $\eta \propto (E_{\text{iso}})^{1/4}$, which is consistent with the prediction of neutrino annihilation from the hyperaccretion disk (for the hot fireball).

For the $\epsilon_\gamma \lesssim 50\%$ case, since $(E_p/E_{\text{ratio}}) \propto (E_{\text{iso}}/E_{\text{ratio}})^{1/4}$ and $\Gamma \propto (L_{\text{iso}}/E_{\text{ratio}})^{1/4}$, we should have $\Gamma \propto (E_p/E_{\text{ratio}})/(T_{90})^{1/4}$. From Figure 9(b), we find this correlation, which is also in line with that for the $\epsilon_\gamma \gtrsim 50\%$ case. Note that we modify E_p or Γ based on Figure 2(d). In Figure 9(c), we show the distribution of $\Gamma - E_p/(T_{90})^{1/4}$ for the large Γ sample (47 bursts) in Ghirlanda et al. (2018). Note that for the

$\epsilon_\gamma \gtrsim 50\%$ case, the E_p is rederived from the $\log(E_p) = 2.54 + 0.25 \log(E_{\text{iso}})$ correlation (using E_{iso}), and for the $\epsilon_\gamma \lesssim 50\%$ case, the E_p is rederived from E_p/E_{ratio} . For the $\epsilon_\gamma \gtrsim 50\%$ case, when calculating Γ , we modify E_k based on the original η/Γ . Obviously, we find that the distribution of Γ and $E_p/(T_{90})^{1/4}$ is centered around $\Gamma = 10^{-0.1} \cdot E_p/(T_{90})^{1/4}$ and shows a linear correlation. Note that the $\Gamma - E_p$ correlation is also found in Ghirlanda et al. (2012).

4.3.3. The Consistency of Our $\Gamma \propto E_{\text{iso}}^{1/8} E_p^{1/2}/(T_{90})^{1/4}$ Correlation and the $\Gamma = 10^{3.33} L_{\text{iso}}^{0.46} E_p^{-0.43}$ Correlation

For the better sample (35 bursts) in Figure 7(d), we replot in Figure 8(a) the comparison of the Γ obtained from the optical afterglow and the Γ obtained from the prompt emission (orange

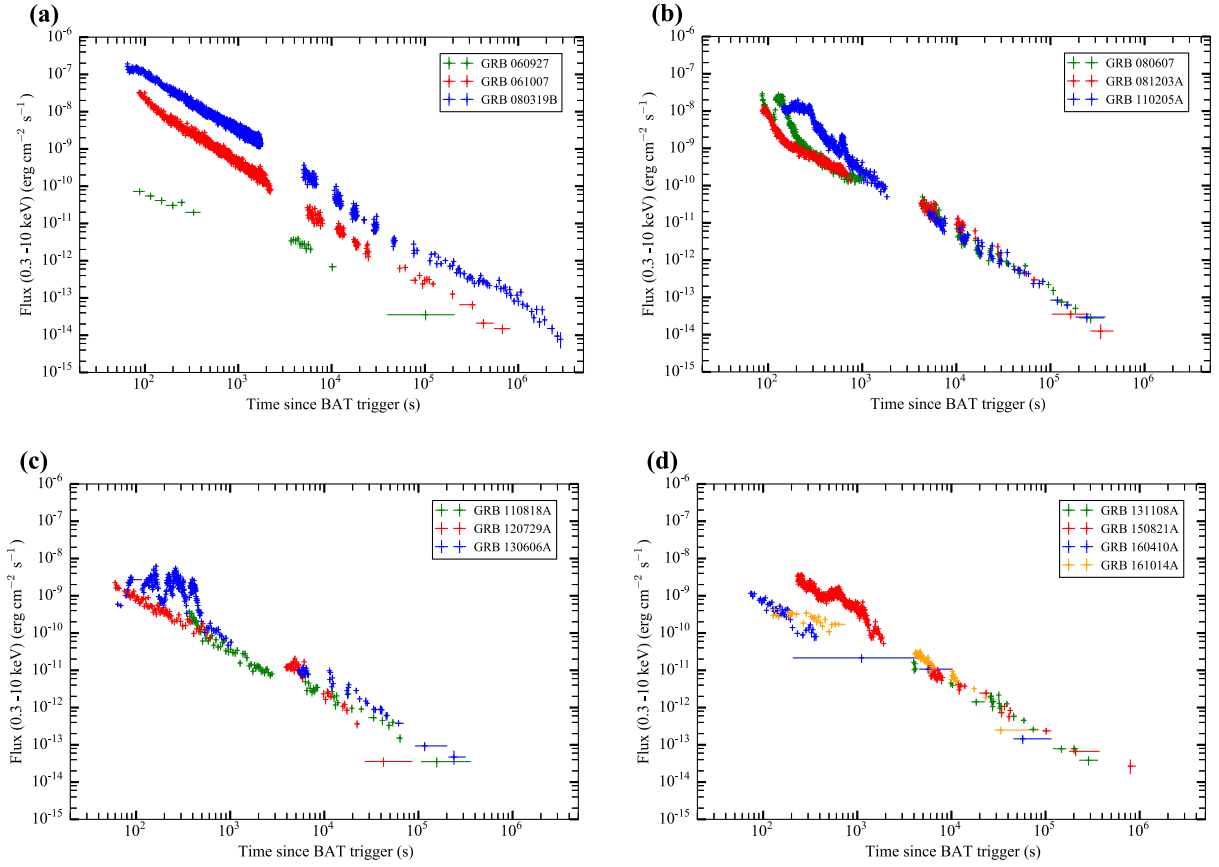


Figure 13. The X-ray afterglow light curves for the selected extremely high-efficiency GRBs (excluding GRB 990705 and GRB 000210). All these light curves appear as simple power-law shapes, without any plateaus, steep decays, or significant flares (with weak flares in the early times).

triangles for $\Gamma = 10^{3.33} L_{\text{iso}}^{0.46} E_p^{-0.43}$ and blue circles for $\Gamma \propto E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4}$. For the $\epsilon_\gamma \gtrsim 50\%$ subsample (17 bursts) and the $\epsilon_\gamma \lesssim 50\%$ subsample (18 bursts), we use the different derived constants. Obviously, Equation (7) as derived in our work gives a much better estimation of Γ (with a much smaller reduced χ^2) compared with Figure 7(c). Noteworthy, the $\Gamma = 10^{3.33} L_{\text{iso}}^{0.46} E_p^{-0.43}$ correlation obtained from the statistical fitting is actually consistent with our $\Gamma \propto E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4}$ correlation derived from the photosphere emission model. Because, along with $E_p \propto (E_{\text{iso}})^{1/4}$ (or $E_p \propto (L_{\text{iso}})^{1/4}$; see Figures 3 and 4), they can be transferred to each other, as shown in the following:

$$\begin{aligned}
 \Gamma &\propto E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4} \\
 &\propto (E_{\text{iso}}/T_{90})^{1/8} \cdot E_p^{1/2} / (T_{90})^{1/8}, \text{ with } E_p \propto (L_{\text{iso}})^{1/4} \\
 &\propto (L_{\text{iso}})^{1/8} \cdot (L_{\text{iso}}^{0.23} \cdot E_p^{-0.43}) / (T_{90})^{1/8} \\
 &\propto L_{\text{iso}}^{0.36} \cdot E_p^{-0.43} / (T_{90})^{1/8}, \text{ with } T_{90} \propto (L_{\text{iso}})^{-0.5} \\
 &\propto L_{\text{iso}}^{0.36} \cdot E_p^{-0.43} / (L_{\text{iso}}^{-0.5})^{1/8} \\
 &\propto L_{\text{iso}}^{0.43} \cdot E_p^{-0.43}.
 \end{aligned} \tag{17}$$

Here, the adopted $T_{90} \propto (L_{\text{iso}})^{-0.5}$ correlation is found from Figure 8(c), for the high-efficiency subsample ($\epsilon_\gamma \gtrsim 80\%$).

In Figure 8(b), we compare the Γ obtained from $\Gamma = 10^{3.33} L_{\text{iso}}^{0.46} E_p^{-0.43}$ and the Γ obtained from $\Gamma \propto E_{\text{iso}}^{1/8} E_p^{1/2} / (T_{90})^{1/4}$ for the $\epsilon_\gamma \gtrsim 50\%$ subsample (17 bursts), the $\epsilon_\gamma \lesssim 50\%$

subsample (18 bursts), and the high-efficiency subsample ($\epsilon_\gamma \gtrsim 80\%$). Obviously, these two estimations are centered around the equal-value line and have linear correlations. Furthermore, for the high-efficiency subsample, which has the tightest $E_p \propto (E_{\text{iso}})^{1/4}$ correlation (with very small dispersion; see Figure 3), these two estimations are almost identical. For the $\epsilon_\gamma \gtrsim 50\%$ subsample and the $\epsilon_\gamma \lesssim 50\%$ subsample, which have larger dispersions for the $E_p \propto (E_{\text{iso}})^{1/4}$ correlation, the above two estimations also show larger dispersions.

4.4. The Distribution of E_{iso}/E_k for the Whole Sample

In Figure 10, we show the distribution of E_{iso}/E_k (indicating the efficiency ϵ_γ) for the whole sample (117 bursts). The average value is around $\sim 10^{-0.2} - 10^{-0.3}$, thus indicating an average efficiency of $\epsilon_\gamma \sim 33\% - 40\%$. From Figure 10, we also find that the distribution seems to consist of three Gaussian distributions. The high-efficiency peak ($\sim 10^{0.6}$) is almost consistent with the $\eta \propto 10^{0.35} \cdot E_p$ and $\Gamma \propto 10^{-0.3} \cdot E_p$ correlations (namely, $E_{\text{iso}}/E_k = \eta/\Gamma \sim 10^{0.65}$) in Figure 9(a). Theoretically, for the $\epsilon_\gamma > 50\%$ case, from Equation (10) and $\eta \propto (L_{\text{iso}})^{1/4}$, we should also have $\Gamma \propto (L_{\text{iso}}/\eta)^{1/3} \propto (L_{\text{iso}}/L_{\text{iso}}^{1/4})^{1/3} \propto (L_{\text{iso}})^{1/4}$. So the high-efficiency peak is predicted to exist. For the $\epsilon_\gamma < 50\%$ case, $E_{\text{iso}}/E_k = (R_{\text{ph}}/R_s)^{-2/3} \propto (L_{\text{iso}}/\Gamma^4)^{-2/3}$. Thus, the low-efficiency peak ($\sim 10^{-0.7}$) is the natural result of $\Gamma \propto (L_{\text{iso}})^{1/4}$, where $\Gamma = \eta$.

Table 8
The Observed Quantities, Inferred Efficiency, and Properties of the Extended Emission for Short GRBs

$\epsilon_\gamma \lesssim 50\%$ Subsample (Eight Bursts)										
GRB	z	$E_{p,z}$ (keV)	E_{iso} (10^{52} erg)	$L_{X,11}$ (10^{45} erg s $^{-1}$)	$E_{\text{iso}} / \left[\frac{3.7 * L_{X,11}}{(1+z) / 2} \right]$	E_{ratio}	$T_{90,i}$ (s)			
051221A	0.5465	$677.0^{+200.0}_{-141.0}$	0.91	0.277	0.687	0.296	0.14			
070724A	0.457	119.5 ± 7.3	0.0016	0.0271	0.0116	0.243	0.27			
070809	2.187	464.0 ± 223.0	0.104	0.338	0.133	0.368	0.44			
130603B	0.356	$823.0^{+83.0}_{-71.0}$	0.196	0.0773	0.465	0.64	0.16			
131004A	0.71	202.0 ± 51.0	0.068	0.145	0.109	0.14	0.9			
140903A	0.351	60.0 ± 22.0	0.0044	0.285	0.00281	0.0691	0.22			
150423A	0.22	146.0 ± 43.0	0.00075	0.0701	0.00176	0.408	1.14			
160821B	0.16	97.4 ± 22.0	0.012	0.0115	0.164	0.0944	0.41			
$\epsilon_\gamma \gtrsim 50\%$ Subsample (Four Bursts)										
GRB	z	$T_{90,i}$ (s)	$L_{X,11}$ (10^{45} erg s $^{-1}$)	$E_{\text{iso}} / \left[\frac{3.7 * L_{X,11}}{(1+z) / 2} \right]$	E_{iso} (10^{52} erg)	$E_{p,z}$ (keV)	$E_{\text{iso}}^{\text{Ext}}$ (10^{52} erg)	$E_{p,z}^{\text{Ext}}$ (keV)	$E_{\text{iso}} / E_{\text{iso}}^{\text{Ext}}$	$E_{p,z} / E_{p,z}^{\text{Ext}}$
070714B	0.923	0.65	0.040	4.127	0.640	1060^{+285}_{-215}	0.116	164.87 ± 73.13	5.517	6.429
110402A	0.805	2.8	0.255	1.454	1.520	1924^{+767}_{-451}	0.642			
150424A	0.30	0.21	0.177	0.431	0.434	1191^{+64}_{-61}	0.0625	82.3 ± 82.1	6.944	14.349
160410A	1.717	0.58	0.441	7.749	9.300	3853^{+1429}_{-973}	1.55	495.3 ± 232.9	6.000	7.779
170817A	0.00968	0.50			$(4.7 \pm 0.7) \times 10^{-6}$	$65.6^{+35.3}_{-14.1}$	$(1.6 \pm 0.2) \times 10^{-6}$	38.4 ± 4.2	3	1.708
061006A	0.4377	0.26			0.382	909^{+260}_{-191}	0.0674	150	5.668	6.060
071227A	0.384	1.30			0.0591	875^{+790}_{-287}	0.0196	80	3.015	10.938
080123A	0.495	0.27			0.32	2228^{+1272}_{-1308}	0.0398	53	8.040	42.038
061210A	0.4095	0.07			0.0024	761^{+648}_{-264}	0.0422			
060614A	0.1254	4.4			0.24	$340^{+241\text{a}}_{-96}$	0.0765			
050709A	0.1606	0.06			0.0027	$96.3^{+20.9\text{a}}_{-13.9}$				
050724A	0.2576	2.4			0.009	$138^{+503\text{a}}_{-57}$				

Note.

^a Only detected by Swift or HETE-2 (lacking detections in the high-energy band).

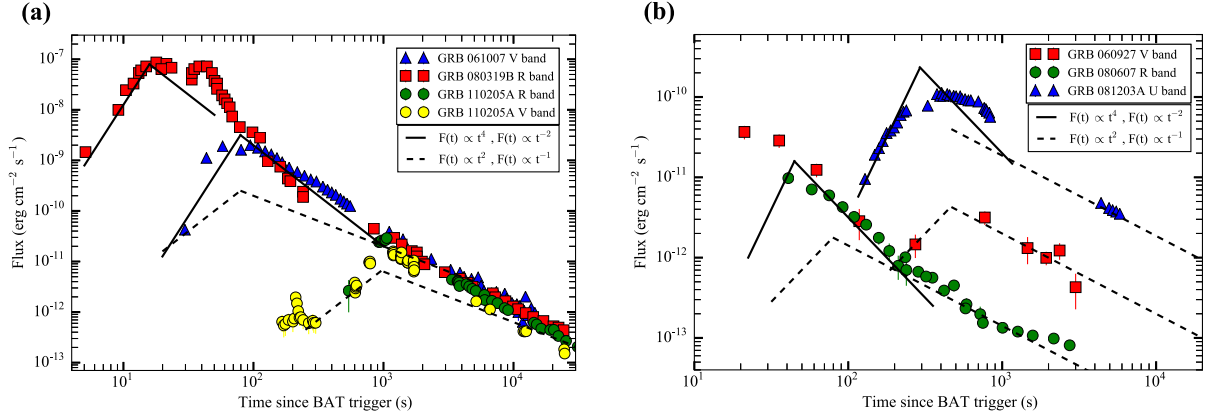


Figure 14. The optical afterglow light curves for six extremely high-efficiency GRBs ($\epsilon_\gamma \gtrsim 80\%$) whose early peaks can be detected. All these light curves show significant reverse shock signals. (a) Three GRBs that are also considered to have reverse shock signals (with detections of both the rapid rise $f \gtrsim t^{-3}$ and the fall $f \sim t^{-2}$) in other works (Gao et al. 2015; Yi et al. 2020). (b) Three GRBs with detections of only the rapid rise $f \gtrsim t^{-3}$ or the rapid fall $f \sim t^{-2}$.

5. Evidence from Short GRBs with $\epsilon_\gamma \gtrsim 50\%$ and $\epsilon_\gamma \lesssim 50\%$

For the short GRBs, similar to the long GRBs, we use the judgment of $E_{\text{iso},52} = E_{k,52} = 3.7^* L_{X,45}$ ($\epsilon_\gamma = 50\%$) to obtain the $\epsilon_\gamma \gtrsim 50\%$ sample (four bursts) and the $\epsilon_\gamma \lesssim 50\%$ sample (eight bursts; see Table 8). In Figure 5(a), we show the E_p and E_{iso} distributions for these two distinguished samples. Obviously, the $\epsilon_\gamma \gtrsim 50\%$ sample and the upmost distribution for the large sample of short GRBs (Zhang et al. 2018b; without $L_{X,45}$ detections for most) follow the $E_p \propto (E_{\text{iso}})^{1/4}$ correlation, which is consistent with the prediction of the photosphere emission model. Noteworthily, the distribution for GRB 170817A well fits the above line ($\log(E_p) = 3.24 + 0.25 \log(E_{\text{iso}})$), too. For the $\epsilon_\gamma \lesssim 50\%$ sample, as predicted, the distribution is below this line (since E_p and E_{iso} are smaller). Similar to Figure 2(a), to test whether E_p and E_{iso} are both smaller by a factor of $E_{\text{iso}}/E_k = (R_{\text{ph}}/R_s)^{-2/3}$, we show in Figure 5(b) the distribution of E_{ratio} ($E_{\text{ratio}} = [(E_p/2.7k)^4 * (4\pi r_1^2 ac)/E_{\text{iso}}]^{1/3}$) and E_{iso}/E_k for this $\epsilon_\gamma \lesssim 50\%$ sample. Note that we exclude three bursts with $E_{\text{iso}} \leq 10^{50}$ erg and that we have $r_1 = 3.4 \times 10^7$ cm here. Again as predicted, they are found to be almost centered around the equal-value line, and have a linear correlation.

Interestingly, we find that all the bursts of the $\epsilon_\gamma \gtrsim 50\%$ sample (five bursts, including GRB 170817A) have extended emission, with the E_p and E_{iso} distribution in Figure 5(a) being for their main pulse. To further test this finding, in Figure 5(c) we show the E_p and E_{iso} distribution of the main pulse for seven other bursts that have extended emission (Minaev & Pozanenko 2020). It is found that, excluding three bursts that are only detected by Swift or HETE-2 (lacking detections in the high-energy band), four other bursts do follow the $\log(E_p) = 3.24 + 0.25 \log(E_{\text{iso}})$ correlation of the $\epsilon_\gamma \gtrsim 50\%$ sample, again supporting the above finding. The true E_p values for these three outliers are likely to be much larger. According to the above, the main pulse for the short GRBs with extended emission is likely to be produced by the photosphere emission in the unsaturated acceleration regime. Then, considering the smaller values of both E_p and E_{iso} for their extended emission, we think that the extended emission may be produced by the transition from unsaturated acceleration to saturated acceleration (E_p and E_{iso} are both smaller by the same factor of $E_{\text{iso}}/E_k = (R_{\text{ph}}/R_s)^{-2/3}$). To test this hypothesis, we show in Figure 5(d) a comparison of the ratios of E_p and the fluence of the main pulse and the extended emission for a large extended

emission sample (including the Swift/BAT bursts with redshift, Gompertz et al. 2020; and the Fermi/GBM bursts without redshift, Lan et al. 2020). As predicted, these two ratios are found to be almost centered around the equal-value line and they have linear correlations.

6. Summary

In this work, after obtaining the prompt emission efficiency of a large GRB sample with redshift, we divide that GRB sample into three subsamples ($\epsilon_\gamma \gtrsim 80\%$, $\epsilon_\gamma \gtrsim 50\%$, and $\epsilon_\gamma \lesssim 50\%$). The well-known Amati relation (Amati et al. 2002) is then well explained by the photosphere emission model. Furthermore, for each subsample, the X-ray and optical afterglow characteristics are consistent with the predictions of the photosphere emission model. Ultimately, large amounts of convincing observational evidence for the photosphere emission model are revealed for the first time.

I thank the anonymous referee for the constructive suggestions. I thank Bin-Bin Zhang and Liang Li for helpful discussions. Y.-Z.M. is supported by the National Postdoctoral Program for Innovative Talents (grant no. BX20200164). This work is supported by the National Key Research and Development Programs of China (2018YFA0404204), the National Natural Science Foundation of China (grant Nos. 11833003, U2038105, 12121003), the science research grants from the China Manned Space Project with No.CMS-CSST-2021-B11, and the Program for Innovative Talents, Entrepreneur in Jiangsu. I also acknowledge the use of public data from the Fermi Science Support Center, Swift, and Konus-Wind.

Appendix A

The Probability Photosphere Model and the Dissipative Photosphere Model

A.1. The Probability Photosphere Model

For the traditional photosphere model, the photosphere emission is all emitted at the photospheric radius R_{ph} , where the optical depth for a photon propagating toward the observer is equal to unity ($\tau = 1$). But if there can be an electron at any position, the photons should have a probability of being scattered there. For an expanding fireball, the photons can be scattered to any place in the fireball with a certain probability. Thus, the traditional spherical shell photosphere is changed to a

probability photosphere, namely the probability photosphere model (Pe'er 2008). Based on careful theoretical derivations, the probability function $P(r, \Omega)$, which gives the probability of a photon being scattered at the radius r and angular coordinate Ω , can be written as (Pe'er 2008; Beloborodov 2011; Lundman et al. 2013)

$$P(r, \Omega) = (1 + \beta)D^2 \times \frac{R_{\text{ph}}}{r^2} \exp\left(-\frac{R_{\text{ph}}}{r}\right), \quad (18)$$

where β is the jet velocity and $D = [\Gamma(1 - \beta \cdot \cos \theta)]^{-1}$ is the Doppler factor.

For the probability photosphere model, the observed photosphere spectrum involves the overlapping of a series of blackbodies with different temperatures, thus its low-energy spectrum is broadened. After considering a jet with angular structure (e.g., Dai & Gou 2001; Zhang & Mészáros 2002; Rossi et al. 2002), the observed typical low-energy photon index $\alpha \sim -1.0$ (Kaneko et al. 2006; Zhang et al. 2011), the spectral evolution, and the E_p evolutions (hard-to-soft evolution or E_p intensity tracking; Liang & Kargatis 1996; Lu et al. 2010, 2012) can be reproduced (Lundman et al. 2013; Meng et al. 2019, 2022).

A.2. The Dissipative Photosphere Model

The dissipative photosphere model (or the subphotosphere model) considers an extra energy dissipation process in the area of moderate optical depth ($1 < \tau < 10$; the subphotosphere). Different dissipative mechanisms have been proposed, such as shocks (Rees & Mészáros 2005), magnetic reconnection (Giannios & Spruit 2007), and proton–neutron nuclear collisions (Vurm & Beloborodov 2016; Beloborodov 2017). Relativistic electrons (with higher temperatures than the photons) are then generated, which upscatter the thermal photons to obtain the nonthermal (broadened) high-energy spectrum.

Appendix B

Some Theoretical Descriptions for the Afterglow

B.1. The Remaining Kinetic Energy in the Afterglow Phase (E_k) and the Isotropic X-Ray Afterglow Luminosity at 11 hr ($L_{X,11h}$)

In the context of the standard afterglow model (Paczynski & Rhoads 1993; Mészáros & Rees 1997), since the X-ray band is above the cooling frequency ν_c at a late afterglow epoch (11 hr), the late-time X-ray afterglow luminosity ($L_{X,11h}$) sensitively depends on only E_k and ϵ_e (the electron equipartition parameter). Furthermore, the fraction of energy in the electrons (ϵ_e) is centered around 0.1, based on afterglow analysis of large samples. Thus, E_k can be well estimated by $L_{X,11h}$ as follows (Lloyd-Ronning & Zhang 2004):

$$\begin{aligned} E_k &= 10^{52} \text{ erg } R \left[\frac{L_{X,11h}}{10^{46} \text{ erg s}^{-1}} \right]^{4/(2+p)} \left(\frac{1+z}{2} \right)^{-1} \\ &\times \epsilon_{e,-1}^{4(1-p)/(2+p)} \epsilon_{B,-2}^{(2-p)/(2+p)} t_{10h}^{(3p-2)/(2+p)} \nu_{18}^{2(p-2)/(2+p)} \\ &\simeq 10^{52} \text{ erg } R \left[\frac{1.1 L_{X,11h}}{10^{46} \text{ erg s}^{-1}} \right] \left(\frac{1+z}{2} \right)^{-1} \\ &\times (\epsilon_{e,-1})^{-1}, \quad (p \simeq 2), \end{aligned} \quad (19)$$

where $R = [t(10 \text{ h})/t(\text{prompt})]^{(17/16)\epsilon_e} \sim 2.27$ constitutes the radiative losses in the first 10 hr after the prompt phase. Note

that the derived E_k is 9.2 times larger in Fan & Piran (2006), since the ν_m (the characteristic frequency corresponding to the minimum electron Lorentz factor) is about one and a half orders smaller. Previously, it had been hard to judge which constant was better. Here, using the method ($\epsilon_\gamma = 50\%$ and the maximum Γ ; see Section 4.1) in this work, our result ($E_{k,52} \simeq 3.7^* L_{X,45}$) is consistent with $E_{k,52} \simeq 2.5^* L_{X,45}$ (Fan & Piran 2006; without the inverse Compton effect).

The isotropic X-ray afterglow luminosity (in the 2–10 keV rest-frame common energy band) at 11 hr (rest frame), $L_{X,11h}$, is computed from the observed integral 0.3–10 keV unabsorbed fluxes at 11 hr ($F_{X,11h}$; estimated from the Swift/XRT light curves) and the measured spectral index Γ_1 (from the XRT spectra), along with the luminosity distance D_L . The equation is as follows (D’Avanzo et al. 2012):

$$L_{X,11h}(2\text{--}10 \text{ keV}) = 4\pi D_L^2 \cdot F_{X,11h}(0.3\text{--}10 \text{ keV}) \cdot \frac{\left(\frac{10}{1+z}\right)^{2-\Gamma_1} - \left(\frac{2}{1+z}\right)^{2-\Gamma_1}}{10^{2-\Gamma_1} - 0.3^{2-\Gamma_1}}. \quad (20)$$

Here, $F_{X,11h}$ is obtained by interpolating (or extrapolating) the best-fit power law for the XRT light curve within a selected time range, including (or close to) 11 hr, for 11 hr.

B.2. The Power-law Shape of the X-Ray Afterglow Predicted by the Classical Hot Fireball

The “generic” afterglow model (relativistic blastwave theory) for GRB predicts a power-law-decaying multiwavelength afterglow (Paczynski & Rhoads 1993; Mészáros & Rees 1997), due to the self-similar nature of the blastwave solution. The observed specific flux is

$$\begin{aligned} f_\nu &\propto E_k^{(p+2)/4} \epsilon_e^{p-1} \epsilon_B^{(p-2)/4} t_{\text{obs}}^{-(3p-2)/4} \nu^{-p/2} \\ &\simeq \frac{(\epsilon_e E_k)}{t_{\text{obs}}} \nu^{-1}, \quad (p \simeq 2). \end{aligned} \quad (21)$$

This power-law behavior ($f \sim t_{\text{obs}}^{-1}$) is consistent with the observations of the optical and radio afterglows. But several surprising emission components (the steep decay $f \sim t_{\text{obs}}^{-3}$ phase, the plateau $f \sim t_{\text{obs}}^{-0.5}$ phase, and the flare) in the early X-ray afterglow are revealed by the Swift observations (Zhang et al. 2006), which are not predicted by the above standard (hot fireball) model. These extra components imply that an extra energy injection (internal or external) may exist, which can be magnetic dominated.

B.3. The Reverse Shock in the Optical Afterglow Predicted by the Classical Hot Fireball

For the classical hot fireball (the magnetic field in the ejecta is dynamically unimportant—namely the magnetization parameter $\sigma \equiv B'^2/(4\pi n'_p m_p c^2) \ll 1$), a strong reverse shock (propagating back across the GRB ejecta to decelerate it) is predicted in the early optical afterglow phase (Mészáros & Rees 1997; Sari & Piran 1999). This prediction is almost confirmed by the discovery of a very bright optical flash in GRB 990123, while the GRB is still active. Later on, many more reverse shock signals are found. The light curve of the reverse shock declines more rapidly ($f \sim t^{-2}$) than that of the

forward shock ($f \sim t^{-1}$), and rises more rapidly ($f \gtrsim t^3$) than that of the forward shock ($f \sim t^2$) before the peak time.

B.4. The Fermi/LAT (GeV Emission) Light Curve from the Radiative Fireball

For the generic afterglow model, the total energy of the fireball remains constant (the adiabatic case) after the forward shock starts to decelerate (entering the self-similar phase). However, there could be another case, in which the total energy of the fireball decreases (the radiative case), since a large fraction of the dissipated energy is radiated away (by magnetic reconnection or electron–proton collisions; Ghisellini et al. 2010). For this radiative fireball, the light curve (after the peak time) declines more rapidly ($f \sim t^{-10/7}$), $\Gamma \propto t^{-3/7}$ ($\Gamma \propto t^{-3/8}$ for the adiabatic case), and the peak time is much earlier ($t_{\text{peak}} = 0.44t_{\text{dec}}$, $t_{\text{peak}} = 0.63t_{\text{dec}}$ for the adiabatic case; t_{dec} is the deceleration time).

ORCID iDs

Yan-Zhi Meng  <https://orcid.org/0000-0002-1122-1146>

References

- Abdo, A. A., Ackermann, M., Ajello, M., et al. 2009, *ApJL*, 706, L138
- Abramowicz, M. A., Novikov, I. D., & Paczynski, B. 1991, *ApJ*, 369, 175
- Acuner, Z., & Ryde, F. 2018, *MNRAS*, 475, 1708
- Acuner, Z., Ryde, F., Pe’er, A., Mortlock, D., & Ahlgren, B. 2020, *ApJ*, 893, 128
- Amati, L., Frontera, F., Tavani, M., et al. 2002, *A&A*, 390, 81
- Axelsson, M., Baldini, L., Barbiellini, G., et al. 2012, *ApJL*, 757, L31
- Axelsson, M., & Borgonovo, L. 2015, *MNRAS*, 447, 3150
- Band, D., Matteson, J., Ford, L., et al. 1993, *ApJ*, 413, 281
- Bégué, D., & Pe’er, A. 2015, *ApJ*, 802, 134
- Beloborodov, A. M. 2011, *ApJ*, 737, 68
- Beloborodov, A. M. 2017, *ApJ*, 838, 125
- Bhattacharya, M., & Kumar, P. 2020, *MNRAS*, 491, 4656
- Bhattacharya, M., Lu, W., Kumar, P., & Santana, R. 2018, *ApJ*, 852, 24
- Burgess, J. M., Bégué, D., Greiner, J., et al. 2020, *NatAs*, 4, 174
- Burgess, J. M., Greiner, J., Bégué, D., & Berlato, F. 2017, arXiv:1710.08362
- Dai, Z. G., & Gou, L. J. 2001, *ApJ*, 552, 72
- D’Avanzo, P., Salvaterra, R., Sbarufatti, B., et al. 2012, *MNRAS*, 425, 506
- Deng, L.-T., Lin, D.-B., Zhou, L., et al. 2022, *ApJL*, 934, L22
- Deng, W., & Zhang, B. 2014, *ApJ*, 785, 112
- Dereli-Bégué, H., Pe’er, A., & Ryde, F. 2020, *ApJ*, 897, 145
- Fan, Y., & Piran, T. 2006, *MNRAS*, 369, 197
- Fan, Y.-Z., Wei, D.-M., Zhang, F.-W., & Zhang, B.-B. 2012, *ApJL*, 755, L6
- Gao, H., Wang, X.-G., Mészáros, P., & Zhang, B. 2015, *ApJ*, 810, 160
- Geng, J.-J., Huang, Y.-F., Wu, X.-F., Zhang, B., & Zong, H.-S. 2018, *ApJS*, 234, 3
- Geng, J.-J., Zhang, B., Kölligan, A., Kuiper, R., & Huang, Y.-F. 2019, *ApJL*, 877, L40
- Ghisellini, G., Ghirlanda, G., Nava, L., & Celotti, A. 2010, *MNRAS*, 403, 926
- Ghirlanda, G., Nava, L., Ghisellini, G., et al. 2012, *MNRAS*, 420, 483
- Ghirlanda, G., Nappo, F., Ghisellini, G., et al. 2018, *A&A*, 609, A112
- Ghirlanda, G., Pescalli, A., & Ghisellini, G. 2013, *MNRAS*, 432, 3237
- Giannios, D., & Spruit, H. C. 2007, *A&A*, 469, 1
- Goldstein, A., Burgess, J. M., Preece, R. D., et al. 2012, *ApJS*, 199, 19
- Gompertz, B. P., Levan, A. J., & Tanvir, N. R. 2020, *ApJ*, 895, 58
- Goodman, J. 1986, *ApJL*, 308, L47
- Gruber, D., Goldstein, A., Weller von Ahlefeld, V., et al. 2014, *ApJS*, 211, 12
- Guiriec, S., Connaughton, V., Briggs, M. S., et al. 2011, *ApJL*, 727, L33
- Guiriec, S., Daigne, F., Hascöet, R., et al. 2013, *ApJ*, 770, 32
- Hou, S.-J., Zhang, B.-B., Meng, Y.-Z., et al. 2018, *ApJ*, 866, 13
- Iyyani, S., & Sharma, V. 2021, *ApJS*, 255, 25
- Kaneko, Y., Preece, R. D., Briggs, M. S., et al. 2006, *ApJS*, 166, 298
- Kobayashi, S., Piran, T., & Sari, R. 1997, *ApJ*, 490, 92
- Lan, L., Lu, R.-J., Lü, H.-J., et al. 2020, *MNRAS*, 492, 3622
- Larsson, J., Racusin, J. L., & Burgess, J. M. 2015, *ApJL*, 800, L34
- Lazzati, D., Morsony, B. J., Margutti, R., & Begelman, M. C. 2013, *ApJ*, 765, 103
- Li, L. 2019a, *ApJS*, 242, 16
- Li, L. 2019b, *ApJS*, 245, 7
- Li, L. 2020, *ApJ*, 894, 100
- Li, L., Geng, J.-J., Meng, Y.-Z., et al. 2019, *ApJ*, 884, 109
- Li, L., Liang, E.-W., Tang, Q.-W., et al. 2012, *ApJ*, 758, 27
- Li, L., Ryde, F., Pe’er, A., Yu, H.-F., & Acuner, Z. 2021, *ApJS*, 254, 35
- Li, L., Wang, Y., Shao, L., et al. 2018, *ApJS*, 234, 26
- Liang, E., & Kargatis, V. 1996, *Natur*, 381, 49
- Liang, E.-W., Li, L., Gao, H., et al. 2013, *ApJ*, 774, 13
- Liang, E.-W., Lin, T.-T., Lü, J., et al. 2015, *ApJ*, 813, 116
- Lin, D.-B., Liu, T., Lin, J., et al. 2018, *ApJ*, 856, 90
- Lloyd-Ronning, N. M., & Zhang, B. 2004, *ApJ*, 613, 477
- Lu, R.-J., Hou, S.-J., & Liang, E.-W. 2010, *ApJ*, 720, 1146
- Lu, R.-J., Wei, J.-J., Liang, E.-W., et al. 2012, *ApJ*, 756, 112
- Lundman, C., Pe’er, A., & Ryde, F. 2013, *MNRAS*, 428, 2430
- Lü, J., Zou, Y.-C., Lei, W.-H., et al. 2012, *ApJ*, 751, 49
- Meng, Y.-Z., Geng, J.-J., Zhang, B.-B., et al. 2018, *ApJ*, 860, 72
- Meng, Y.-Z., Liu, L.-D., Wei, J.-J., Wu, X.-F., & Zhang, B.-B. 2019, *ApJ*, 882, 26
- Meng, Y.-Z., Geng, J.-J., & Wu, X.-F. 2022, *MNRAS*, 509, 6047
- Mészáros, P., & Rees, M. J. 1997, *ApJ*, 476, 232
- Mészáros, P., & Rees, M. J. 2000, *ApJ*, 530, 292
- Mészáros, P. 2002, *ARA&A*, 40, 137
- Minaev, P. Y., & Pozanenko, A. S. 2020, *MNRAS*, 492, 1919
- Nava, L., Salvaterra, R., Ghirlanda, G., et al. 2012, *MNRAS*, 421, 1256
- Paczynski, B. 1986, *ApJL*, 308, L43
- Paczynski, B., & Rhoads, J. E. 1993, *ApJL*, 418, L5
- Parsotan, T., & Lazzati, D. 2022, *ApJ*, 926, 104
- Pe’er, A. 2008, *ApJ*, 682, 463
- Pe’er, A., Barlow, H., O’Mahony, S., et al. 2015, *ApJ*, 813, 127
- Pe’er, A., & Ryde, F. 2011, *ApJ*, 732, 49
- Piran, T. 1999, *PhR*, 314, 575
- Rees, M. J., & Meszaros, P. 1994, *ApJL*, 430, L93
- Rees, M. J., & Mészáros, P. 2005, *ApJ*, 628, 847
- Rossi, E., Lazzati, D., & Rees, M. J. 2002, *MNRAS*, 332, 945
- Ruffini, R., Siutsou, I. A., & Vereshchagin, G. V. 2013, *ApJ*, 772, 11
- Ryde, F. 2004, *ApJ*, 614, 827
- Ryde, F. 2005, *ApJL*, 625, L95
- Ryde, F., Axelsson, M., Zhang, B. B., et al. 2010, *ApJL*, 709, L172
- Ryde, F., Lundman, C., & Acuner, Z. 2017, *MNRAS*, 472, 1897
- Ryde, F., & Pe’er, A. 2009, *ApJ*, 702, 1211
- Sari, R., & Piran, T. 1999, *ApJ*, 520, 641
- Song, X., & Meng, Y.-Z. 2022, *MNRAS*, 512, 5693
- Tang, Q.-W., Wang, K., Li, L., & Liu, R.-Y. 2021, *ApJ*, 922, 255
- Thompson, C. 1994, *MNRAS*, 270, 480
- Tsvetkova, A., Frederiks, D., Golenetskii, S., et al. 2017, *ApJ*, 850, 161
- Uhm, Z. L., & Zhang, B. 2014, *NatPh*, 10, 351
- Vereshchagin, G. V., & Siutsou, I. A. 2020, *MNRAS*, 494, 1463
- Vereshchagin, G., Li, L., & Bégué, D. 2022, *MNRAS*, 512, 4846
- von Kienlin, A., Meegan, C. A., Paciesas, W. S., et al. 2020, *ApJ*, 893, 46
- Vurm, I., & Beloborodov, A. M. 2016, *ApJ*, 831, 175
- Vyas, M. K., Pe’er, A., & Eichler, D. 2021, *ApJ*, 908, 9
- Wang, K., Lin, D.-B., Wang, Y., et al. 2020, *ApJ*, 899, 111
- Wang, Y., Zheng, T.-C., & Jin, Z.-P. 2022, arXiv:2205.08427
- Wygodna, N., Guetta, D., Mandich, M. A., & Waxman, E. 2016, *ApJ*, 824, 127
- Xue, L., Zhang, F.-W., & Zhu, S.-Y. 2019, *ApJ*, 876, 77
- Yamazaki, R., Sato, Y., Sakamoto, T., & Serino, M. 2020, *MNRAS*, 494, 5259
- Yang, J., Chand, V., Zhang, B.-B., et al. 2020, *ApJ*, 899, 106
- Yi, S.-X., Wu, X.-F., Zou, Y.-C., & Dai, Z.-G. 2020, *ApJ*, 895, 94
- Yu, H.-F., Preece, R. D., Greiner, J., et al. 2016, *A&A*, 588, A135
- Yu, H.-F., van Eerten, H. J., Greiner, J., et al. 2015, *A&A*, 583, A129
- Zhang, B. 2011, *CRPhy*, 12, 206
- Zhang, B. 2020, *NatAs*, 4, 210
- Zhang, B., Fan, Y. Z., Dyks, J., et al. 2006, *ApJ*, 642, 354
- Zhang, B., Liang, E., Page, K. L., et al. 2007, *ApJ*, 655, 989
- Zhang, B., & Mészáros, P. 2002, *ApJ*, 571, 876
- Zhang, B., & Yan, H. 2011, *ApJ*, 726, 90
- Zhang, B.-B., Liu, Z.-K., Peng, Z.-K., et al. 2021, *NatAs*, 5, 911
- Zhang, B.-B., Zhang, B., Liang, E.-W., et al. 2011, *ApJ*, 730, 141
- Zhang, B.-B., Zhang, B., Castro-Tirado, A. J., et al. 2018a, *NatAs*, 2, 69
- Zhang, B.-B., Zhang, B., Sun, H., et al. 2018b, *NatCo*, 9, 447
- Zhang, Z. J., Zhang, B.-B., & Meng, Y.-Z. 2021, arXiv:2109.14252
- Zhao, P.-W., Tang, Q.-W., Zou, Y.-C., & Wang, K. 2022, *ApJ*, 929, 179